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Can real-time investor sentiment help predict the high-frequency stock returns? Evidence from a mixed-frequency-rolling decomposition forecasting method

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ABSTRACT

This research examines the predictive effect of real-time investor sentiment on high-frequency stock returns. Utilizing text sentiment analysis, we extract investor sentiment with a half-hour frequency from the stock message board. The RR-MIDAS method is used to model half-hourly sentiment and three-minute stock returns, and economic analysis reveals that investor sentiment significantly affects the stock returns during seven high-frequency periods, and the influence gradually weakens. Subsequently, we propose the “MF-EEMD-ML” prediction system, which introduces a rolling decomposition algorithm into the RR-MIDAS framework for predicting high-frequency trend items combined with real-time forum sentiment. The results, using rolling EMD decomposition for comparison, show that the “MF-EEMD-ML” system achieves a maximum reduction of 19.18 % in MAE, 19.08 % in RMSE, 11.71 % in SMAPE, and a maximum improvement of 16.66 % in DS. Additionally, the outcomes of the Diebold-Mariano (DM) tests also demonstrate that the “MF-EEMD-ML” prediction system significantly outperforms both the “MF-EMD-ML” system and the LR model.

1. Introduction

The Efficient Market Hypothesis (EMH) holds that asset prices have incorporated all historical information in a weak-form efficient market, so predicting stock price trends solely based on historical data is unfeasible. However, Behavioral Finance focuses on

Abbreviations: VAR, Vector Autoregression; OLS, Ordinary Least Square; AIC, Akaike Information Criterion; ARIMA, Autoregressive Integrated Moving Average; QR, Quantile Regression; SVR, Support Vector Regression; RF, Random Forest; XGBoost, eXtreme Gradient Boosting Regression; ANN, Artificial Neural Networks; RNN, Recurrent Neural Network; LSTM, Long Short-Term Memory; GRU, Gated Recurrent Unit; CNN, Convolutional Neural Network; BiLSTM, Bidirectional Long Short-Term Memory; BERT, Bidirectional Encoder Representations from Transformers; MSE, Mean Squared Error; RMSE, Root Mean Squared Error; SMAPE, Symmetric Mean Absolute Percentage Error; MAPE, Mean Absolute Percentage Error; DS, Direction precision of statistics; R^2 , Coefficient of Determination; AUC, Area Under the Curve; RU-MIDAS, Reverse Unrestricted Mixed-Data Sampling; RR-MIDAS, Reverse Restricted Mixed-Data Sampling; TEI@I, Text mining + Econometrics + Intelligence @ Integration; VMD, Variational Mode Decomposition; EMD, Empirical Mode Decomposition; EEMD, Empirical Ensemble Mode Decomposition; MF-EMD-ML, Mixed Frequency- Empirical Mode Decomposition-Machine Learning; MF-EEMD-ML, Mixed Frequency, Empirical Ensemble Mode Decomposition -Machine Learning; DM test, Diebold-Mariano test.

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individual behavior and psychological factors, aiming to forecast asset prices by analyzing deviations and anomalies in individual performance (Shiller, 2003). Empirical evidence indicates that investor sentiment can predict asset prices. For instance, Chung, Hung, and Yeh (2012) discover that investor sentiment holds predictive power primarily during economic expansions rather than contractions, and Antoniou, Doukas, and Subrahmanyam (2016) note that sentiment-driven trading increases during emotionally charged periods. Investor sentiment notably affects stock return patterns and market anomalies (Shen, Liu, & Zhang, 2018).

The impact of investor sentiment on capital markets has prompted substantial research (Ding & Hou, 2015; Dimpfl & Jank, 2016; Han, Li, & Yin, 2018; Wen, Xu, Ouyang, & Kou, 2019). Three primary methods can measure investor sentiment: the survey method (Schmeling, 2009; Ciner, 2014; Greenwood & Shleifer, 2014), proxy variables method (Chen, Wu, Jin, & Wang, 2021; Cevik, Altinkeski, Cevik, & Dibooglu, 2022), and big data analysis (Li, Bu, Li, & Wu, 2020; Agoraki, Aslanidis, & Kouretas, 2022; Swamy & Lagesh, 2023). The survey method is costly and challenging for continuous tracking, offering weaker prediction power than market indicators (Zhou, 2018). Single variables, such as mutual fund flow (Frazzini & Lamont, 2008), stock turnover (Wang & Hu, 2022), and the VIX index (Hoekstra & Guler, 2022), lack comprehensive representation. Furthermore, the composite index, such as the investor sentiment constructed by Baker and Wurgler (2006) through principal component analysis, lacks consistency in selecting underlying indicators across different scenarios, thus making it difficult to compare conclusions.

With the maturation of big data and internet technology, accessing multi-source information from online platforms and extracting public opinions have emerged as prominent research trends (Ashtiani & Raahmei, 2023). Major text sources for exploring the linkage between investor sentiment and the stock market include news reports (Allen, McAleer, & Singh, 2019; Fraiberger, Lee, Puy, &

Table 1
Summary of existing researches.

Panel A Research on the relationship between textual data from different sources and financial markets				
Author	Study	Main data source	Empirical approach	Main conclusion
Costola et al. (2023)	Machine learning sentiment analysis, COVID-19 news and stock market reactions	MarketWatch.com, The New York Times, and Thomson Reuters	Multivariate VAR type models	sentiment is positively correlated with market returns
Fraiberger et al. (2021)	Media sentiment and international asset prices	Newspaper published by Reuters	Multivariate VAR type models	Global news sentiment has a larger impact on returns
Chang et al. (2022)	Can stock message board sentiment predict future returns?	guba.eastmoney.com of China	OLS	Predictability of small-cap stock returns comes from local online postings
Choi and Choi (2019)	Effects of limited attention on investors' trading behavior	Daum, the 2nd largest Internet portal in Korea	QR, the CAPM model, and Fama-French model	Investor attention is associated with an increase in trading volume
Leung and Ton (2015)	The impact of internet stock message boards on cross-sectional returns of small-capitalization stocks	Messages covers over 2000 companies listed on the Australian Securities Exchange	Event study, fixed effect panel regressions, and CAPM model	Number of message board is positively correlated with the returns and trading volume of small-cap stocks
Li et al. (2023)	Understanding the role of social media sentiment in identifying irrational herding behavior in the stock market	200,000 pieces of textual data from Sina Weibo	Econometric approach	Social media sentiment exacerbates herding behavior in the stock market
Miwa (2023)	Divergent opinions on social media	Bloomberg's social sentiment analytics	Cross-sectional regressions	Social media divergent opinions can influence subsequent changes in analyst target prices and quarterly earnings forecasts
Gu and Kurov (2020)	Informational role of social media	Bloomberg's firm-specific Twitter sentiment	Fama-MacBeth type regression	Twitter sentiment has one-third predictability for stock returns
Panel B Research on predictions using machine learning and text sentiment analysis				
Author	Market	Methodology	Main features	Performance metric
Jin et al. (2020)	Chinese Industry stock index	SVR	Stock market variables and sentiment in online comments	MSE, MAPE, and DS
Gupta, Nel, and Pierdzioch (2023)	Dow Jones Industrial Average	RF	Monthly realized volatility and investor confidence index	RMSE, R^2 , and Clark-West Test
Deng et al. (2023)	the Shanghai composite index and Shenzhen composite index	XGBoost	Sentiment features of institutional, individual and foreign investors	Accuracy, F1-score, and accumulated return
Chiong, Fan, Hu, and Dhakal (2022)	Germany stock market	RNN, LSTM, and GRU	Financial news sentiment and historical stock data	Accuracy, precision, recall, F1-score, and AUC
Eslamieh, Shajari, and Nickabadi (2023)	Dow Jones Industrial Average	CNN-BiLSTM	Vector of social media sentiment and stock market data	Accuracy, precision, recall, and F1-score

Ranciere, 2021; Costola, Hinz, Nofer, & Pelizzon, 2023), stock message boards (Leung & Ton, 2015; Choi & Choi, 2019; Chang, Shao, & Wang, 2022), and social media platforms (Bartov, Faurel, & Mohanram, 2018; Broadstock & Zhang, 2019; Li, Chen, Liu, Yu, & Yu, 2023). Uhl and Novacek (2021) employ rolling correlation analysis between Thomson Reuters news themes and the S&P 500 index, establishing headline sentiment variables that reveal a significant impact on stock prices. Chang et al. (2022) find that the predictability of small-cap stock returns is mainly driven by local posts on China's largest stock message board, guba.eastmoney.com. Similarly, Miwa (2023) computes company-specific sentiment scores using Twitter and StockTwits, demonstrating that dissenting social media opinions can predict future stock returns. Table 1, Panel A, outlines research exploring the link between investor sentiment from various text sources and financial markets.

The impact of social media on financial markets is profound due to the generation of ample unstructured user opinions. Scholars extensively use text information to build models that predict future asset prices. Machine learning algorithms, which can discern patterns between features and labels, have been widely applied in the modeling and forecasting of financial time series (Long, Lu, & Cui, 2019; Deng et al., 2023; Hao, Yuan, Wu, Xu, & Li, 2023). These algorithms, including SVR, XGBoost, LSTM, and CNN, are mainly used for predicting financial asset prices. For example, Jin, Guo, Sun, Lai, and Liao (2020) extract investor sentiment from social media platforms and compare the forecasting efficacy of combining this sentiment or not combining it. They find that the SVR model, when combined with investor sentiment, outperforms ARIMA and ANN models in predictive performance. Deng et al. (2024) utilize institutional, individual, and foreign investors sentiment as input features, and combine multiple XGBoost models with the Bagging method. Their empirical findings indicate that this approach yields high accuracy, low risk, and stable returns in predicting trends in Chinese stock index prices and in simulating trading. Lu, Liu, and Chen (2021) create a hybrid-frequency investor sentiment measure by combining micro and macro sentiments. Their empirical results suggest that the ANN model's ability to predict stock market crises significantly surpasses that of six other benchmark models. Table 1, Panel B, summarizes research that involves predictions using machine learning and text sentiment analysis.

The target variables in the aforementioned studies generally pertain to two areas: the binary classification problem of predicting the rise and fall of stocks, or forecasting stock index values. However, forecasting stock returns is widely recognized as more challenging. This task demands not only accurately discerning the direction of stock movements but also quantifying these movements. Zhao and Yang (2023) have highlighted the difficulties in achieving precise forecasts due to the amplified noise and clustered volatility in stock return data. To address this, the TEI@I complex system approach, proposed by Wang, Yu, and Lai (2005), proves beneficial. It decomposes raw data into components of varying time scales, predicts each component individually, and then integrates these predictions to obtain the final fitted value. The TEI@I method has enhanced prediction accuracy and has been successfully applied in forecasting carbon, electricity, and crude oil prices (Sun & Zhang, 2018; Wang, Luo, Grunder, Lin, & Guo, 2017; Zhang, Tang, Wu, Du, & Chen, 2021). This method has also been used in financial asset forecasting. For example, Gao and Zhang (2023) apply the VMD algorithm to decompose investor sentiment and stock returns into short-term, medium-term, and long-term trends. They combine the sentiment and forecast the return components of different time scales separately. Empirical evidence suggests that this hybrid model significantly reduces errors and enhances predictive performance when incorporating investor sentiment.

Unfortunately, Yang, Wang, Niu, and Du (2020) point out a data leakage issue in existing predictive studies using the TEI@I method. This issue arises when the entire dataset is first decomposed, then divided into training and testing sets for fitting and forecasting. Such an approach raises concerns. Specifically, in the training set perspective, the decomposed components in the testing set remain unknown due to the connection between training and testing sets through historical and future relationships. Since future data cannot be known in advance, it is impossible to predict it beforehand. Yang et al. (2020) emphasize that utilizing a rolling decomposition and one-step-ahead prediction approach is the correct methodology. However, the adoption of this rolling decomposition algorithm is currently limited in the field of predictive research.

This study aims to explore whether real-time sentiment from stock message boards can improve stock return prediction and to develop a specialized model for heightened accuracy. We have gathered over a million posts from the largest Chinese stock message board, utilizing SnowNLP sentiment analysis to assign sentiment polarity to the posts. Unlike previous research, our interest lies in the predictive role of intraday investor sentiment for high-frequency stock returns. While obtaining minute-level stock price data is straightforward, forum posts may have intermittent gaps and limited posting quantity at this level, representing only a portion of real-time opinions. To address this, we extract emotional indicators from the message board every half hour. Extracting forum sentiments at half-hour intervals not only addresses the issue of missing posts in certain minutes but is also more beneficial to achieve the research objectives of this paper. The aim is to discuss the dynamic changes in the impact of real-time social media sentiments on high-frequency stock returns. This frequency has been acknowledged in the studies of both Sun, Najand, and Shen (2016) and Yin et al. (2022). However, it presents a challenge in mixed-frequency modeling between investor sentiment and stock returns. The Reverse Unrestricted Mixed-Data Sampling model (RU-MIDAS), introduced by Foroni, Guerin, and Marcellino (2018), offers a solution. It enables the integration of low-frequency data in predicting high-frequency variables by performing frequency alignment on the high-frequency data over different time periods.

Inspired by Foroni's (2018) concept of reverse mixed-frequency data sampling, we focus on 3-minute stock returns and divide the data into ten subseries periods. To address the challenge of directly predicting stock returns in conjunction with message board sentiment, we employ a decomposition algorithm for each subsequence. Specifically, the EEMD algorithm is employed for decomposition, and fuzzy entropy values are computed for the decomposed components, allowing us to reconstruct them into high-frequency, medium-frequency, and low-frequency components. Our analysis reveals that the high-frequency components show the lowest sequence autocorrelation, making them more suitable for forecasting combined with message board sentiment. In contrast, medium-frequency and low-frequency components rely solely on their historical information for prediction. The final prediction is obtained by linearly integrating the results from the high-frequency, medium-frequency, and low-frequency components. To evaluate

the effectiveness of the EEMD decomposition algorithm, this paper initially explores the empirical mode decomposition algorithm (EMD). Moreover, to ensure stable prediction results, we use six machine learning algorithms—SVR, RF, XGB, LSTM, GRU, and CNN—for predictive modeling. Essentially, our primary contributions involve:

(1) Utilizing data mining techniques, we gather a substantial number of posts from stock message boards. We obtain the investor sentiment indicator through both manual labeling and automatic sentiment scoring using SnowNLP analysis. This text-based sentiment index captures investor sentiments more effectively than survey-based and proxy variable methods. However, our study focuses on the dynamic impact of intraday high-frequency investor sentiment on the stock market. Extracting sentiment from message boards at minute-level intervals reflects only a small portion of investor viewpoints and may encounter periods with missing posts. To address this, we extract investor sentiment at half-hourly intervals. We then use the RR-MIDAS model to establish a relationship between half-hourly investor sentiment and 3-minute stock returns. This approach allows for a more genuine and accurate representation of the dynamic impact generated by intraday high-frequency investor sentiment.

(2) The RR-MIDAS results show a significant impact of half-hourly sentiment on 3-minute stock returns during seven high-frequency periods. Following an economic impact analysis, we proceed to construct a forecasting model. Due to the non-stationary

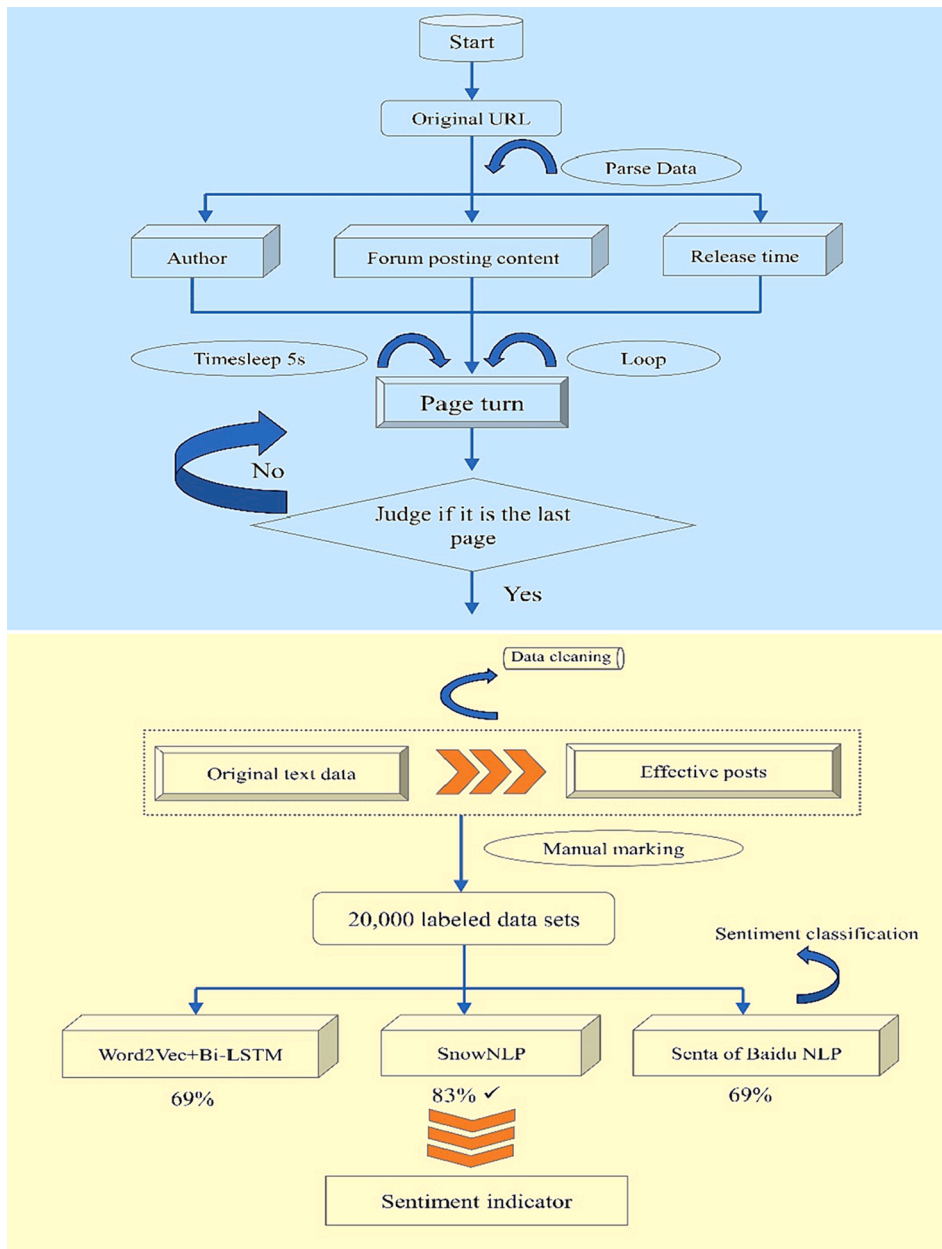


Fig. 1. Process of text mining and sentiment analysis.

and nonlinear nature of stock returns, we integrate the rolling EEMD decomposition algorithm into the RR-MIDAS model. This approach, as discussed in Yang et al. (2020) study, addresses the common issue of data leakage in decomposition forecasting research. Rolling decomposition effectively mitigates this problem. We apply this technique to each subseries of 3-minute stock returns, in line with the RR-MIDAS approach. After decomposition, we reassemble the components into three distinct time-scale categories based on fuzzy entropy values. The high-frequency components, characterized by low sequence autocorrelation, are combined with half-hourly investor sentiment for prediction using machine learning algorithms. Conversely, medium-frequency and low-frequency components rely solely on historical information for forecasting. We aggregate predictions from these three components to produce the final fitted value. The window moves forward one step at a time, repeating the process until the last data point. This method, termed “MF-EEMD-ML”, draws inspiration from the RU-MIDAS model. It not only integrates half-hourly message board sentiment, which significantly influences the stock market, but also employs decomposition algorithms to handle complex data.

(3) In the decomposition phase, as outlined in point (2), our research uses the EMD decomposition method as a benchmark to assess the effectiveness of the EEMD algorithm. To ensure the stability of our prediction results and to avoid bias towards any specific machine learning algorithm, we select six classic algorithms. These include three from traditional machine learning and three from deep learning. We consider the forecasting system “MF-EEMD-ML” effective when most algorithms show consistent advantages in their forecasting results. Finally, to confirm the robustness of our findings, we perform the Diebold-Mariano test on all prediction results.

The remaining sections of this study are arranged as follows: Part Two introduces the construction of stock message boards sentiment, the Reverse Unrestricted Mixed Data Sampling Model, the EEMD method, and the establishment of the “MF-EEMD-ML” prediction framework. Part Three presents the results of descriptive statistical analysis, the dynamic impact of half-hourly message board sentiment on 3-minute stock returns, and the sequence decomposition, fuzzy entropy computation, and sequence reconstruction processes. Part Four initially demonstrates the inadequate performance of non-decomposed mixed-frequency prediction algorithms, followed by an analysis of the results of the “MF-EMD-ML” and “MF-EEMD-ML” prediction frameworks. Finally, Part Five provides a comprehensive summary of the entire study.

2. Data and model

2.1. Stock message board posts and sentiment measure

To achieve real-time dynamic forecasting of intraday investor sentiment on stock returns, the selection of a representative source of textual data is essential. In China, small and medium-sized investors, who often express their opinions on stock message boards, form the largest group of stockholders. This study specifically focuses on the Shanghai Composite Index forum on Eastmoney Finance, chosen for its high posting frequency, substantial content, and strong traceability. The Eastmoney guba forum has similarly been used as a textual data source in studies by Gao, Xiong, Feng, Li, and Vigne (2019) and Huang, Wen, Yang, Cao, and Yang (2022).

We employ web crawling to gather content from stock forums, as depicted in the left half of Fig. 1. The data fields we collect include the author, posting content, and posting time. To circumvent anti-crawling mechanisms on websites, we introduce a 5-second delay before each page turn and execute the process in a loop. The original posts contain irrelevant information and require cleaning. The main cleaning criteria involve: (1) removing posts with only images or advertisements; (2) deleting blank posts; (3) eliminating reposts from other financial forums; (4) excluding posts about news, institutional research reports, and company announcements. After cleaning, we obtain 1,068,183 valid posts, spanning from January 16, 2020, to November 18, 2022, within the trading hours of 9:30–11:30 and 13:00–15:00. In Section 2.4, during the “MF-EEMD-ML” predictive modeling process, we face the challenge of computational burden due to rolling window decomposition and machine learning techniques. To manage this, we adopt a random sampling approach, selecting 10,000 data entries from April 27, 2022, at 13:33 to November 1, 2022, at 15:00.

Combining manual annotation with machine learning is the predominant approach in recent studies for classifying financial texts (Uhl & Novacek, 2021; Liu, Zhang, Qiao, & Xu, 2022). The lower portion of Fig. 1 illustrates this sentiment classification process. We randomly select 20,000 posts for manual labeling by five research students experienced in the stock market. Posts are labeled as “1” for positive, “0” for negative, and ambiguous posts are later labeled by the advisor. We experiment with three algorithms: (1) Text vectorization using Word2Vec and BiLSTM training; (2) Sentiment analysis using the SnowNLP Bayesian machine learning algorithm; (3) Sentiment analysis using BaiduNLP. Examination results of 1000 out-of-sample comments indicate that method (2) performs best, achieving an accuracy of 83 %. Shi et al. (2019) combine text vector representation and SVM algorithm to train manually labeled forum posts, achieving an accuracy of 76 %. Yin et al. (2022) supplement multiple sources of sentiment lexicons and use a model consistent with our study, achieving a classification accuracy of 86 %. Therefore, it can be considered that the accuracy of SnowNLP used in this study for classifying stock forum posts is satisfactory.

We calculate the quantity of positive and negative posts every thirty minutes and construct the investor sentiment indicator Sen_t :

$$Sen_t = \ln\left(\frac{1 + N_t^{buy}}{1 + N_t^{sell}}\right) \quad (1)$$

In Formula (1), N_t^{buy} represents the count of positive posts during the specified interval, and N_t^{sell} represents the count of negative posts. When $N_t^{buy} > N_t^{sell}$, $Sen_t > 0$, indicating that investors are optimistic during that period; conversely, if $N_t^{buy} < N_t^{sell}$, Sen_t reflects a pessimistic sentiment.

2.2. Reverse unrestricted mixed data sampling (RU-MIDAS)

Foroni et al. (2018) propose the RU-MIDAS model, which could utilize the low-frequency data to predict high-frequency variables. Define $\{Sen_t, t \in Z\}$ as the half-hourly frequency investor sentiment sequence, the delay interval can be denoted as $L^L Sen_t = Sen_{t-1}$, and L^L denotes the lag operator of $\{Sen_t, t \in Z\}$. Define $\{return_\tau, \tau \in Z\}$ as the three-minute interval stock returns, its delay interval can be represented as $L^H return_\tau = return_{\tau-1/m}$, L^H denotes the lag operator of $\{return_\tau, \tau \in Z\}$, and the frequency multiplier difference between sentiment and returns is 10. In the neighboring time intervals of t and $t+1$ within the half-hourly investor sentiment series, there are 10 three-minute stock returns labeled as $return_{t+1/10}$, which occur at discrete time points $\tau = t + 1/10, t + 2/10, \dots, t + 1$, where h ranges from 1 to 10.

We use the half-hourly sentiment indicator, Sen_t , to predict the three-minute target variable, $return_\tau$. This process involves aligning and converting $return_\tau$ into a low-frequency format. We achieve this by using the frequency multiplier difference ‘10’. Consequently, the low-frequency format is represented as the vector $(return_{t+1/10}, return_{t+2/10}, \dots, return_{t+10/10})^T$, $t = 1, 2, \dots, T$. This transformation results in the formation of a corresponding matrix, which is detailed in equation (2).

$$return_\tau = \begin{bmatrix} return_{1/10} & return_{2/10} & \cdots & return_{10/10} \\ return_{1+1/10} & return_{1+2/10} & \cdots & return_{1+10/10} \\ \cdots & \cdots & \cdots & \cdots \\ return_{t+1/10} & return_{t+2/10} & \cdots & return_{t+10/10} \\ \cdots & \cdots & \cdots & \cdots \\ return_{T-1+1/10} & return_{T-1+2/10} & \cdots & return_{T/10} \end{bmatrix} \quad (2)$$

Given the frequency difference of 10 between high-frequency stock returns and low-frequency investor sentiment, the reverse mixed data sampling method constructs 10 models. This allows for the analysis of the dynamic linkage between these two variables and enables the prediction of target variables. The function of the RU-MIDAS model can be expressed as follows:

$$return_{t+h/m} = \alpha_{h,0} + \sum_{j=1}^l \alpha_{h,j} return_{t+h/m-j/m} + \sum_{i=0}^p \beta_{h,i} Sen_{t-i} + \varepsilon_{t+h/m} \quad (3)$$

In equation (3), $\alpha_{h,0}$ represents the intercept term, p and l denote the maximum lag order of investor sentiment series and stock returns sequence, $\beta_{h,i}$ and $\alpha_{h,j}$ represent the regression coefficients of the sentiment and stock returns, $\varepsilon_{t+h/m}$ is the stochastic disturbance term, and $return_{t+h/m}$ is the m_{th} high-frequency sequence in the adjacent half-hourly interval.

2.3. Ensemble Empirical Mode decomposition (EEMD)

The EMD method is more effective in handling non-stationary and nonlinear data patterns compared to traditional approaches. It has the capability to decompose complex signals into various Intrinsic Mode Functions (IMFs) and a residual. These IMFs and residuals collectively capture the local characteristic of original sequence at different time scales. Theoretically, it holds the potential to decompose time series of varying properties. However, the EMD algorithm may face mode mixing issues. To overcome this, the Ensemble Empirical Mode Decomposition (EEMD) method introduces white noise into the original sequence before decomposition. The ultimate modes are then obtained by averaging these decomposed components. The specific steps of the EEMD process are outlined below:

(1) Define the total added times of white noise as N , for each iteration, Gaussian white noise $n_i(t)$ is added to the original signal $x(t)$, creating distinct signal-to-noise ratio mixed components $x_i(t)$:

$$x_i(t) = x(t) + n_i(t) \quad (4)$$

(2) Apply the Empirical Mode Decomposition (EMD) method to each mixture component $x_i(t)$ after the addition of Gaussian white noise. This process yields some intrinsic mode functions $IMF_{ij}(t)$ and a residual term $r_i(t)$, which can be represented by Equation (5):

$$x_i(t) = \sum_{j=1}^m IMF_{ij}(t) + r_i(t) \quad (5)$$

(3) After iterating through (1) and (2) N times, calculate the average of the respective components to mitigate the influence of the added white noise:

$$IMF_j(t) = 1 \left/ N \sum_{i=1}^N IMF_{ij}(t) \right. \quad (6)$$

For a comprehensive explanation of the EMD method, refer to Rezaei, Faaljou, and Mansourfar (2021).

2.4. Modeling of “MF-EEMD-ML” system

In this section, we provide a detailed overview of the Mixed Frequency-Rolling Ensemble Empirical Mode Decomposition-Machine Learning model, abbreviated as “MF-EEMD-ML”. The frequency alignment procedures for this system have already been discussed in

Section 2.2. Step 4 of the process involves six machine learning algorithms: SVR, RF, XGB, LSTM, GRU, and CNN. Due to space constraints, we do not provide specific details about each algorithm here. For more information, readers can refer to the studies by Jin et al. (2020), Fazlija and Harder (2022), Deng et al. (2023), Zolfaghari and Gholami (2021), Jiang, Hu, Xiao, and Dong (2022), and Ma

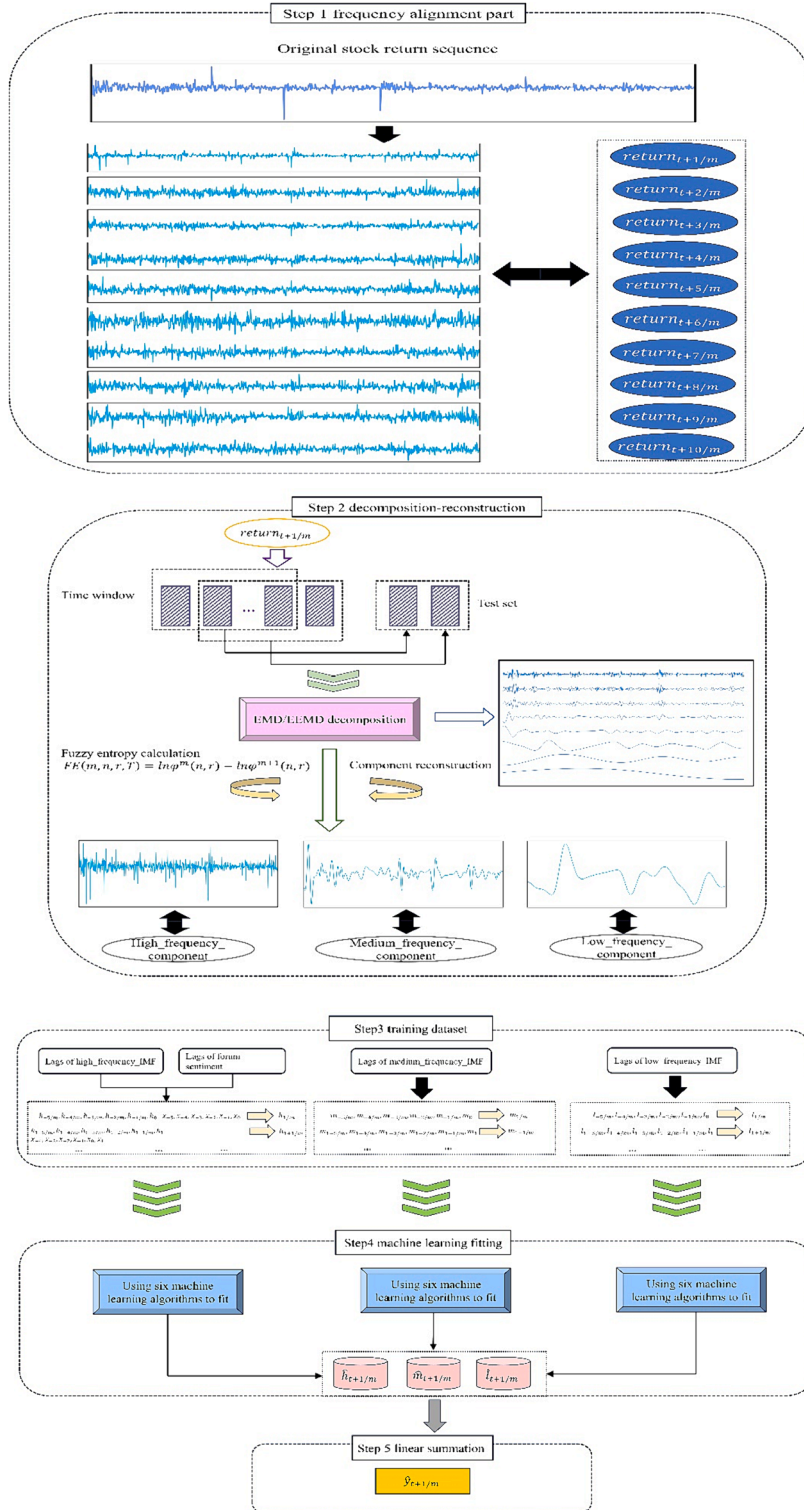


Fig. 2. Modeling of the “MF-EEMD-ML” prediction system.

and Yan (2022). Fig. 2 illustrates the entire modeling process.

Step 1: Forum sentiment is recorded at half-hour intervals, while stock index returns are captured every 3 min. The first step, therefore, involves aligning the return series. We start with the return corresponding to the initial time point. Then, every 10th interval is sampled to create the first return subsequence $\{return_{t+1/10}\}$. The corresponding time points for this subsequence are $\{2022042709 : 33, 2022042710 : 03, \dots, 2022110114 : 33\}$. Similarly, the second to tenth subsequences are generated, resulting in a complete sequence ranging from $\{return_{t+2/10}\}$ to $\{return_{t+10/10}\}$.

Fig. 2 Modeling of the “MF-EEMD-ML” prediction system.

Step 2: We employ a fixed window size of 800, subjecting each subsequence to rolling decomposition. This initial decomposition typically yields 8 to 10 distinct frequency components, as illustrated in the flowchart (Fig. 2). For each component, we calculate the fuzzy entropy values and then aggregate components with similar values for reconstruction. This process results in three trend components: high-frequency, mid-frequency, and low-frequency, as detailed in the procedure. For more information on fuzzy entropy calculation and component reconstruction, refer to Chen, Vivian, and Ye (2022).

Step 3: The construction of the training dataset involves defining lag periods for the high-frequency trend components identified in Step 2. We select lag periods of 5 for the high-frequency, medium-frequency, and low-frequency components by applying the AIC criterion to various subsequences within the first fixed window of the training set. Similarly, a lag period of 5 is determined for the sentiment indicator associated with high-frequency items, resulting in a lagged sequence $\{h_{t-5/m}, h_{t-4/m}, \dots, h_{t-1/m}, h_t\}$. We also establish corresponding half-hour forum sentiment sequences $\{Sen_{t-5}, Sen_{t-4}, \dots, Sen_{t-1}, Sen_t\}$ for these lag periods. For the mid-frequency and low-frequency sequences, lag periods are established only for their respective components, without incorporating sentiment variables.

Step 4: For the datasets corresponding to the high-frequency, mid-frequency, and low-frequency trends, we divide them into training and test sets with ratios of 0.8 and 0.2, respectively. These sets are then sequentially used as inputs for fitting six different machine learning algorithms. It is important to note that we further partition the initial 0.8 training set into sub-training and validation sets, using ratios of 0.8 and 0.2 respectively. This is done during the initial decomposition of each sequence. Optimal parameters, determined based on the results from the validation set, are then used to refit the original 0.8 training set. Table 2 displays the hyperparameters of the six models for the first sub-sequence. After completing this step, we obtain one-step-ahead fitting values for the three trend components, represented as $\{\hat{h}_{t+1/m}, \hat{m}_{t+1/m}, \hat{l}_{t+1/m}\}$.

Step 5: We calculate the predicted value for $return_{t+1/m}$ by summing the fitted values of the three trend components. This study utilizes rolling decomposition to tackle the data leakage issue commonly found in previous research. The process outlined in Step 2 involves moving the fixed window forward a short distance and then repeating Steps 2 through 4. This cycle continues until we reach the final data point in the dataset, thereby completing the “mixed-frequency - rolling decomposition - one-step ahead prediction” procedure.

The pseudocode for the modeling process mentioned above is shown in Fig. 3.

2.5. Evaluation metrics

In this study, RMSE, MAE, SMAPE, and DS are used as indicators to evaluate the predictive performance. The calculation formulas for these indicators are shown in equations (7) to (10):

$$RMSE = \sqrt{1/T \sum_{t=1}^T (y_t - \hat{y}_t)^2} \quad (7)$$

$$MAE = 1/T \sum_{t=1}^T |y_t - \hat{y}_t| \quad (8)$$

$$SMAPE = 1/T \sum_{t=1}^T |y_t - \hat{y}_t| / \left(\frac{|y_t| + |\hat{y}_t|}{2} \right) \quad (9)$$

Table 2

Hyperparameters of the six models in the first sub-sequence.

Model	Hyperparameters
SVR	C:10, epsilon:0.05, gamma:'scale'
RF	n_estimators:200, max_depth:5, max_leaf_nodes:100
XGB	n_estimators:100, max_depth:7, learning_rate:0.12
LSTM	LSTM (units:64, activation:'relu') + Dropout (0.25) + Dense
GRU	GRU (units:128, activation:'relu') + Dropout (0.25) + Dense
CNN	Conv1D (filters:32, kernel_size:3, activation:'relu') + MaxPooling1D (pool_size:2) + Flatten + Dense

Note: All of these hyperparameters are configured within the “Keras” and “Tensorflow” packages of Python. Hyperparameter values for the second through the tenth subsequences are available upon request.

Algorithm 1: Modeling of the "MF-EEMD-ML" system.

Data: Forum sentiment at half-hour intervals, stock index returns every 3 minutes.
Result: Predictive values for stock return subsequences.

- 1 **Step 1: Frequency Alignment;**
- 2 Align the return series by sampling every 10th interval to create the first return subsequence $\{\text{return_}(t+1/10)\}$;
- 3 **for each subsequent interval do**
- 4 Generate subsequences leading to a complete sequence from $\{\text{return_}(t+2/10)\}$ to $\{\text{return_}(t+10/10)\}$;
- 5 **end**
- 6 **Step 2: Decomposition and Reconstruction;**
- 7 **for each subsequence with a fixed window size of 800 do**
- 8 Subject subsequence to rolling EEMD decomposition;
- 9 Calculate fuzzy entropy for each decomposed component;
- 10 Reconstruct components into high-frequency, mid-frequency, and low-frequency trends;
- 11 **end**
- 12 **Step 3: Training Dataset Construction;**
- 13 **for each trend component (high, mid, low-frequency) do**
- 14 Define lag periods of 5 using AIC criterion;
- 15 Create lagged sequences for stock returns and half-hour forum sentiment;
- 16 **end**
- 17 **Step 4: Machine Learning Fitting;**
- 18 **for datasets corresponding to high, mid, low-frequency trends do**
- 19 Divide into training and test sets with ratios of 0.8 and 0.2 respectively;
- 20 Partition training set into sub-training and validation sets with ratios of 0.8 and 0.2 respectively;
- 21 **for each machine learning algorithm (SVR, RF, XGB, LSTM, GRU, CNN) do**
- 22 Optimize parameters based on validation set;
- 23 Refit algorithm on training set using optimized parameters;
- 24 Predict one-step-ahead values for trend components;
- 25 **end**
- 26 **end**
- 27 **Step 5: Linear Summation;**
- 28 **for each predicted timestep $t+1/m$ do**
- 29 Calculate the predicted value by summing the fitted values of the trend components;
- 30 Move the fixed window forward and repeat steps 2 through 4 until reaching the last data point;
- 31 **end**

Fig. 3. Pseudocode for the "MF-EEMD-ML" modelling process.

$$DS = 1 / \left(n \sum_{t=1}^n a(t) a(t) \begin{cases} 1, y_t \hat{y}_t \geq 0 \\ 0, \text{else} \end{cases} \right) \quad (10)$$

In equations (7) to (10), T is the length of samples in the test set, where y_t is the true value and $\hat{y}_t$ is the fitted value.

3. Statistical descriptions and economic analysis

3.1. Descriptive statistics

In Section 2.1, we utilize text mining and sentiment analysis techniques to create an intraday half-hourly investor sentiment index. Fig. 4 illustrates both the investor sentiment index and the first stock return subsequence derived from mixed-frequency sampling. Table 3 provides a concise summary of their descriptive statistics. As shown in Fig. 4(a), the investor sentiment index predominantly

displays negative values and exhibits pronounced fluctuations. This suggests that users on the discussion board mainly express pessimistic emotions, a finding supported by the mean and the 25th and 75th percentile values in Table 3. In contrast, the stock returns subsequence depicted in Fig. 4(b) shows relatively stable volatility. This observation is corroborated by the descriptive statistics in Table 3, which indicate that the stock returns fluctuate around zero. Notably, the kurtosis of the stock returns is 16.42, reflecting the “heavy tails” characteristic commonly seen in financial time series data.

3.2. Dynamic impact analysis using RR-MIDAS

In this context, we apply the RR-MIDAS model, introduced by Xu, Zhuo, Jiang, Sun, and Huang (2021), to analyze the dynamic impact of half-hourly frequency sentiment on subsequent 3-minute frequency stock returns over ten distinct periods. The RR-MIDAS model addresses the “curse of dimensionality”, which arises from significant frequency disparities and excessive lag periods, by incorporating a polynomial weighting function for both stock returns and investor sentiment. The modeling equation for RR-MIDAS, presented in Equation (11), uses consistent symbol definitions as established in Equation (3). In this model, α_h and β_h represent common parameters across lag periods for both stock returns and lagged sentiment on the discussion board. The variation in coefficients across different lag periods is determined by the weight functions $\omega(\delta_h; j)$ and $\omega(\delta_h; i)$. This polynomial weighting function employs the Almon weight function, formulated in Equation (12). For our implementation, we use a single-parameter Almon function with an initial δ of -0.5 , indicating that the influence decreases as the lag period increases. Fig. 5 shows the variation in weight values for five lagged periods when δ is set to -0.5 . The model employs a non-linear least squares estimation (NLS) approach.

$$return_{t+h/m} = \alpha_{h,0} + \alpha_h \sum_{j=1}^l \omega(\delta_h; j) return_{t+h/m-j/m} + \beta_h \sum_{i=0}^p \omega(\delta_h; i) Sen_{t-i} + \varepsilon_{t+h/m} \quad (11)$$

$$\omega(\delta, j) = e^{\delta_1 j + \delta_2 j^2 + \dots + \delta_q j^q} / \sum_{j=1}^l e^{\delta_1 j + \delta_2 j^2 + \dots + \delta_q j^q} \quad (12)$$

Table 4 presents the estimation results, while Fig. 5 shows the coefficient variations in the impact of message board sentiment, recorded at half-hourly frequency, on 3-minute stock returns during ten high-frequency periods. The coefficient variations of α_h in Table 4 indicate that even the lagged periods of the high-frequency returns series struggle to produce significant effects. A notable exception is during the $h = 8$ period, where a positive influence is observed with a coefficient value of 0.2077. Examining the β_h column in Table 4 reveals that half-hourly sentiment significantly impacts future 3-minute stock returns in seven high-frequency periods:

$h = 2, 4, 5, 7, 8, 9, 10$. From the coefficient variations shown in Fig. 6, it can be discerned that the impact of investor sentiment on stock returns gradually diminishes amid fluctuations. Specifically, the coefficient rises from -0.0198 to 0.0222 between $h = 2$ and $h = 4$ periods, and from -0.0271 to 0.0168 between $h = 7$ and $h = 8$ periods.

Empirical results of Sun et al. (2016) demonstrate that the social media investor sentiment, lagged by half an hour, can predict intraday returns of the S&P 500 Index. Moreover, this influence is independent of the index’s own momentum effects. The economic analysis in this study indicates that intraday forum sentiment in China has a more significant impact on high-frequency stock returns than the returns themselves. More importantly, this research uses the RR-MIDAS model to analyze the effect of half-hour forum sentiment on minute-by-minute stock index returns, highlighting the dynamic nature of emotional impacts. However, previous studies use the identical-frequency data, and fail to capture this dynamic process.

3.3. Sequence decomposition, fuzzy entropy calculation, and reconstruction

In the first rolling window of the initial stock returns subsequence, we use the EEMD algorithm to decompose the original stock returns sequence into various modal components of distinct frequencies, as illustrated in Fig. 7. After this decomposition, we obtain a total of seven Intrinsic Mode Functions (IMFs) and one residual component, with these components being ordered based on their

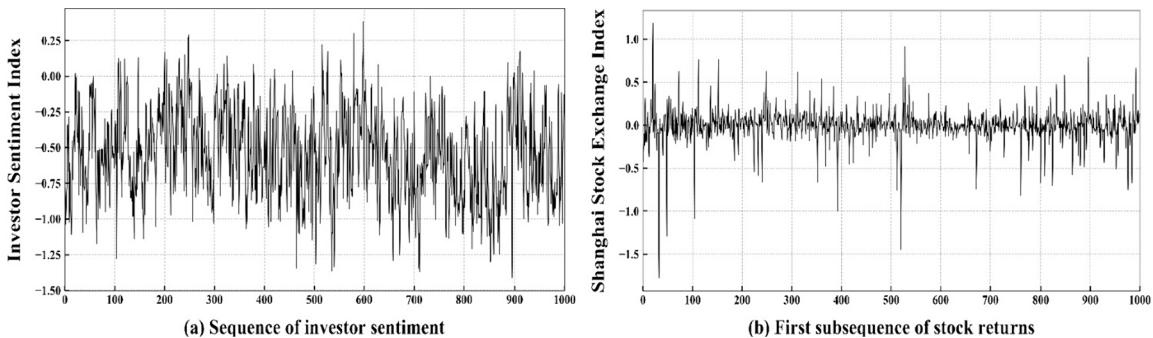
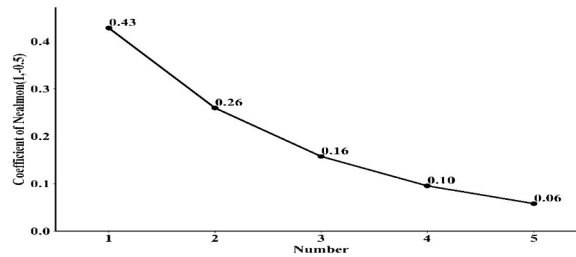


Fig. 4. Graphs of stock message board sentiment and high-frequency stock returns.

Table 3

Statistical description of sentiment and first subsequence of stock returns.

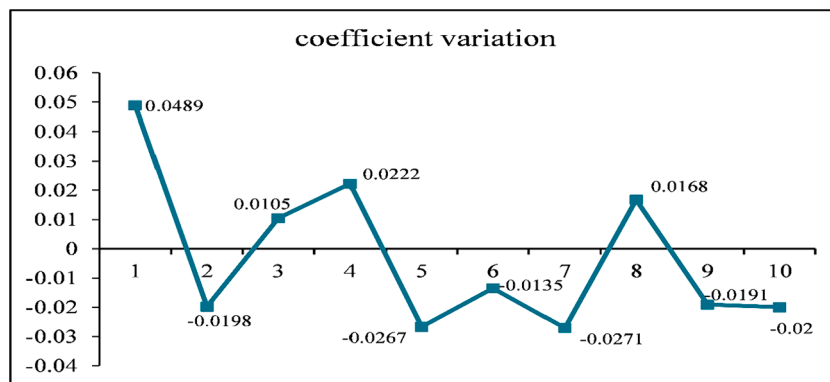
	Mean	First Quantile	Median	Third Quantile	Maximum	Minimum	Std	Skewness	Kurtosis
<i>Sen</i>	−0.54	−0.79	−0.54	−0.32	0.38	−1.41	0.32	0.05	−0.50
<i>Return</i>	0	−0.07	0	0.07	1.19	−1.78	0.20	−1.52	16.42

**Fig. 5.** Coefficients of Almon weight function with δ taking values of -0.5 .**Table 4**

RR-MIDAS regression results.

Period	$\alpha_{h,0}$	$return_{t+h/m-j/m}$		Sen_{t-i}	
		α_h	δ_h	β_h	δ_h
$h = 1$	0.0246 (0.2280)	−0.2356 (0.2130)	−0.0048 (0.9810)	0.0489 (0.1610)	0.0820 (0.8350)
$h = 2$	−0.0049 (0.4492)	−0.0426 (0.3355)	0.8897 (0.7285)	−0.0198* (0.0934)	1.7083 (0.6571)
$h = 3$	0.0039 (0.6670)	0.0468 (0.5790)	−1.1275 (0.6620)	0.0105 (0.4990)	0.8037 (0.7460)
$h = 4$	0.0061 (0.3294)	0.0610 (0.4255)	−1.0102 (0.5482)	0.0222** (0.0454)	−3.6199 (0.8403)
$h = 5$	−0.0086 (0.2642)	0.1131 (0.3315)	0.3452 (0.5175)	−0.0267* (0.0527)	0.5828 (0.3366)
$h = 6$	−0.0020 (0.7560)	−0.1329 (0.2690)	0.0698 (0.7870)	−0.0135 (0.2180)	1.3192 (0.6270)
$h = 7$	−0.0159* (0.0628)	−0.0363 (0.6932)	−0.1400 (0.9114)	−0.0271* (0.0887)	0.4294 (0.4278)
$h = 8$	0.0049 (0.4032)	0.2077*** (0.0021)	0.4404 (0.3279)	0.0168* (0.0999)	4.4829 (0.9317)
$h = 9$	−0.0093 (0.1634)	0.1117 (0.1019)	−1.2844 (0.4155)	−0.0191* (0.0876)	1.6109 (0.5509)
$h = 10$	−0.0129** (0.0238)	0.0545 (0.3852)	1.0052 (0.6730)	−0.0200** (0.0434)	6.8859 (0.9884)

Note: Values outside parentheses are coefficient values, and values inside parentheses are p-values. ***, **, and * respectively indicate significance at the 1 %, 5 %, and 10 % confidence levels.

**Fig. 6.** Coefficient variations of stock message board sentiment impact on high-frequency stock returns.

descending frequency.

After obtaining the decomposed components, we calculate the fuzzy entropy, with the results displayed in Fig. 8. These results show a decreasing trend in fuzzy entropy from IMF1 to IMF8. Based on the decomposition outcomes from Fig. 7, we reconstruct IMF1 and IMF2 as high-frequency components, IMF3, IMF4, and IMF5 as mid-frequency elements, and IMF6, IMF7, and IMF8 as low-frequency trends. This reconstruction process is detailed in Equation (13). Fig. 9 illustrates the reconstructed components: the high-frequency component shows pronounced volatility, the mid-frequency portion displays moderated fluctuations, and the low-frequency

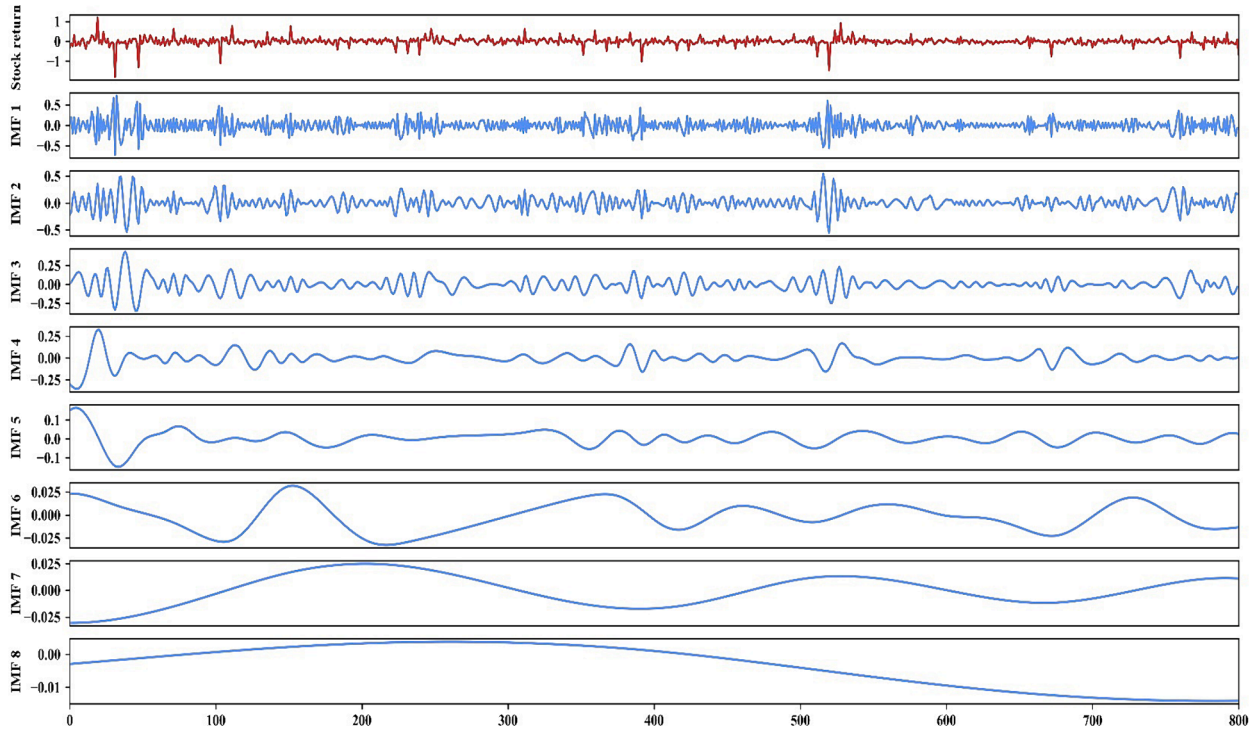


Fig. 7. Stock returns and the decomposed components.

segment captures the long-term trend variations in stock returns.

$$C_H(t) = \sum_{j=1}^2 C_j(t), C_M(t) = \sum_{j=3}^5 C_j(t), C_L(t) = \sum_{j=6}^8 C_j(t) \quad (13)$$

For the high-frequency, mid-frequency, and low-frequency components, we implement a 5-lag period approach. Using Pearson correlation coefficients, we calculate the correlation between each sequence and its corresponding lagged sequence. The results are presented in Table 5. It can be observed that the low-frequency component demonstrates a strong correlation with its own lagged series over 5 periods, with correlation coefficients nearly reaching 1 and being statistically significant. Conversely, the mid-frequency component shows an increasing trend in correlation with its own lagged series as the period decreases. However, the high-frequency component does not exhibit a substantial correlation with its own lagged series. Even at lagged periods 2 and 1, the correlations are only -0.14 and -0.20 , respectively. Therefore, in the prediction phase, we only incorporate the half-hourly investor sentiment as an auxiliary variable for the high-frequency component. The mid-frequency and low-frequency components, meanwhile, achieve desirable outcomes through the utilization of their own lagged periods.

4. Forecasting of 3-minute stock returns

4.1. Forecasting without decomposition algorithm

To demonstrate the efficacy of the proposed “MF-EEMD-ML” framework, we initially explore the predictive outcomes of the RU-

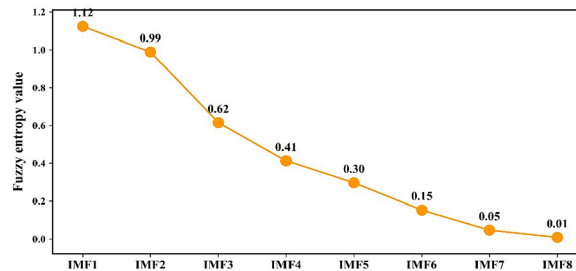


Fig. 8. Fuzzy entropy values of the eight decomposed components.

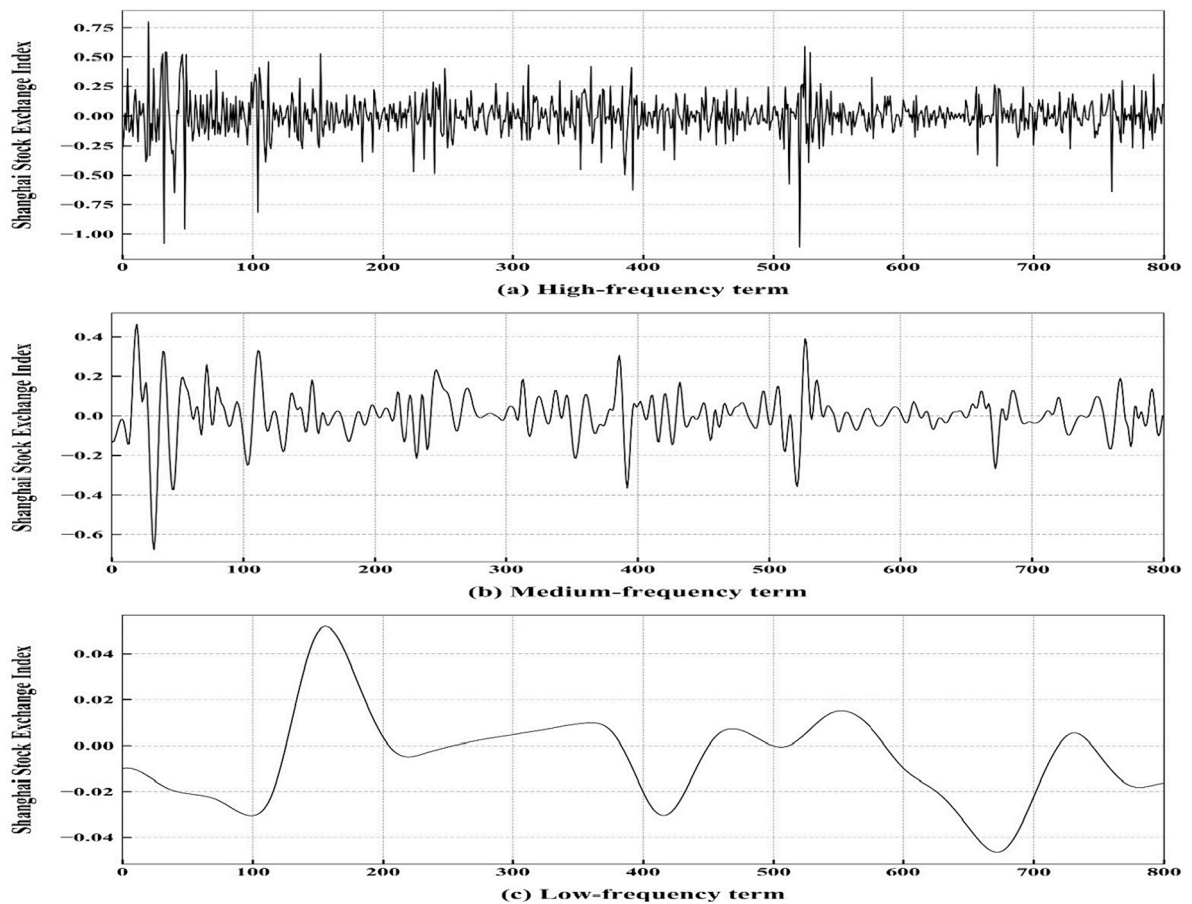


Fig. 9. Three reconstructed components.

Table 5

Pearson correlation coefficient results.

	5-period lag	4-period lag	3-period lag	2-period lag	1-period lag
<i>H_IMF</i>	−0.01 (0.67)	0 (0.91)	−0.04 (0.26)	−0.14*** (8.31e-05)	−0.20*** (1.59e-08)
<i>M_IMF</i>	−0.20*** (2.45e-08)	0.03 (0.35)	0.34*** (3.60e-23)	0.66*** (1.27e-101)	0.91*** (1.67e-301)
<i>L_IMF</i>	0.98***(0)	0.99***(0)	0.99***(0)	1.00***(0)	1.00***(0)

Note: The values above represent Pearson correlation coefficients, and the values below correspond to their respective p-values. ***, **, and * respectively indicate significance at the 1 %, 5 %, and 10 % confidence levels.

MIDAS approach without decomposition. This approach combines half-hourly investor sentiment with machine learning algorithms for forecasting. Fig. 10(a)–(d) illustrates the results. In terms of Mean Absolute Error (MAE), only LSTM (0.1294) and GRU (0.1282) achieve values lower than the LR model (0.1299). For Root Mean Square Error (RMSE), SVR (0.1975), LSTM (0.1994), and GRU (0.1972) outperform the LR model (0.2014). In the case of Symmetric Mean Absolute Percentage Error (SMAPE), SVR (1.4873), RF (1.4852), XGB (1.4675), and CNN (1.5009) record lower values than the LR model (1.5851). Additionally, regarding the Directional precision of Statistics (DS), all six models surpass the LR model. However, merely evaluating the level and vertical error metrics is not sufficient to confirm the effectiveness of forgoing decomposition and relying solely on mixed-frequency machine learning models combined with investor sentiment for forecasting.

Fig. 10(e) presents a time series plot that showcases the predictive outcomes of all models. The blue line represents the actual stock returns, and lines in various colors indicate the fitted values from six mixed-frequency machine learning algorithms. These predictions appear to lack clear patterns and discernible trends. Additionally, their fluctuations are modest, which suggests an inability to accurately fit the actual trends. Table 6 includes the Diebold-Mariano (DM) test results, where the t-statistics for the six machine learning algorithms are collectively non-significant. This suggests that, within the mixed-frequency forecasting framework and without the use of decomposition algorithms, the machine learning models fail to demonstrate their anticipated advantages over linear

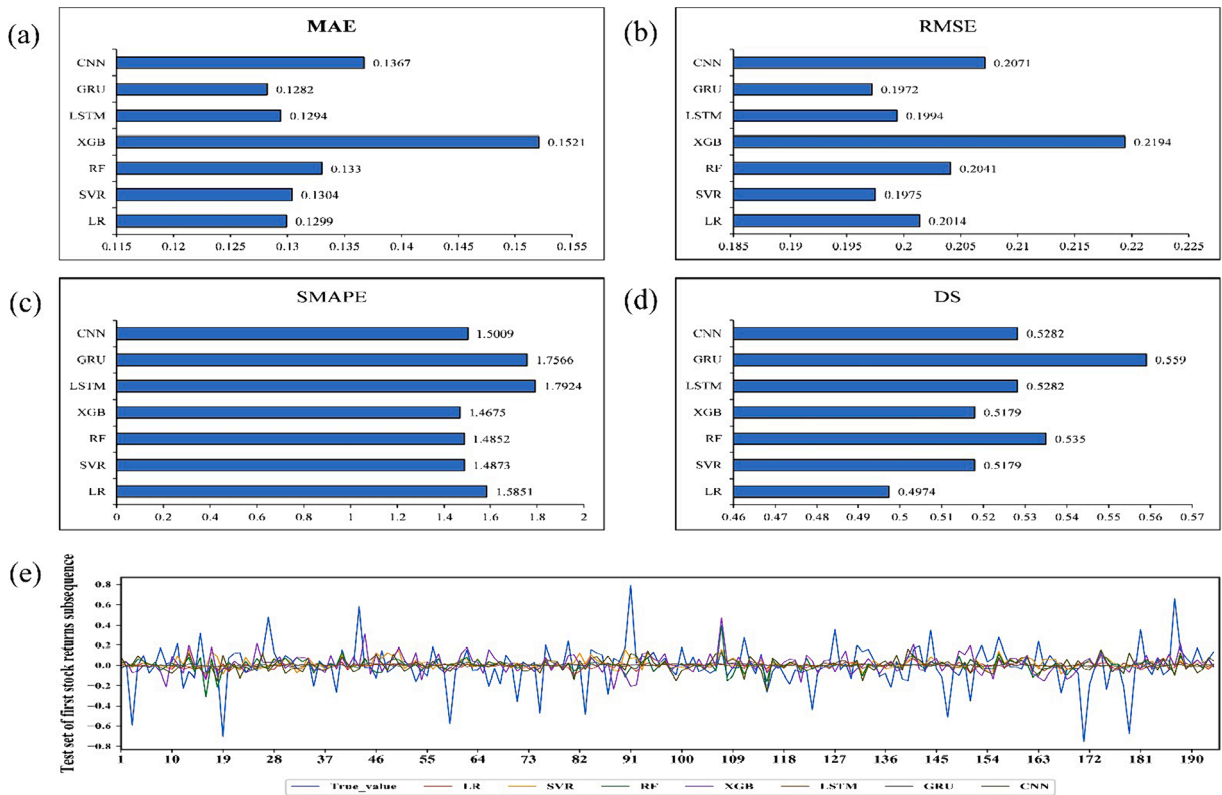


Fig. 10. Prediction results of the first sub-sequence without the decomposition algorithm.

regression. Contrasting with the studies of Long et al. (2019), who predict the direction of stock price movements, and Deng, Zhu, Yu, and Huang (2024), who forecast stock price indices, this research finds that even with the utilization of real-time investor sentiment from forums, precise prediction of stock returns remains a challenge. Although there is a minor improvement in error indicators, forecasting without a decomposition algorithm still falls short of achieving accurate trend prediction fitting.

In summary, although we incorporated half-hourly investor sentiment from stock message boards in this section, the unsatisfactory predictive performance of the reverse mixed-frequency model was likely due to the inherent high noise and clustered volatility in the stock returns sequence. The predictive outcomes for the second to tenth subsequences of stock returns, which underwent mixed-frequency periodic treatment, showed patterns similar to the first subsequence, leading the six algorithms to underperform compared to the LR model. Therefore, Fig. 10 and Table 6 only display the predictive outcomes for the first sub-series of stock returns in the undecomposed framework, while the results for the remaining nine sub-series are available upon request.

4.2. Forecasting results of the “Mixed-Frequency-Rolling Decomposition-Prediction” framework

4.2.1. Prediction results of the “MF-EMD-ML” system

This section details the predictive outcomes of the proposed “Mixed-Frequency-Rolling Decomposition-Forecasting” framework employed in this study, which utilizes the EMD algorithm for decomposition. Appendix Table A1 outlines the horizontal and vertical accuracy indicators for all the predictive results of the model. Fig. 11, in the form of a bar chart, illustrates the degree of improvement for each model over the original linear regression (LR) model. This comparison is made across four metrics and considers the ten sub-series of stock index returns that have undergone reverse mixed-frequency treatment.

Fig. 11 shows that, except for the MAE indicators in the first, seventh, and eighth sub-series, and the RMSE indicator in the eighth sub-series where some models demonstrate growth, the remaining predictive results exhibit improvements over the LR model.

Table 6

DM test results (compared to LR model) under the non-decomposition framework.

	SVR	RF	XGB	LSTM	GRU	CNN
t-statistic	−0.0931	−0.9304	−4.2021***	0.2672	0.8737	−1.6877*
p value	(0.9259)	(0.3522)	(2.6448e-05)	(0.7893)	(0.3823)	(0.0915)

Note: Performance of the model is considered distinct from the LR model only when t-statistic is positive and the p-value is less than 0.1.

Notably, significant enhancements are seen in the tenth sub-series predictions: the MAE indicator for the MF-EMD-CNN model has been reduced by 17.21 %, and the RMSE indicator for the MF-EMD-GRU model has decreased by 22.52 %. Additionally, in the predictions for the second sub-series, the MF-EMD-LSTM model's SMAPE indicator has decreased by 35.25 %, and the DS indicator for the MF-EMD-SVR model has improved by 41.13 %.

We have conducted Diebold-Mariano (DM) tests as a subsequent step, with the outcomes documented in the appended [Table A2](#). In this table, each column represents a baseline model, and each row corresponds to a target model. Our findings indicate that, although the “MF-EMD-ML” prediction framework performs well on four types of error metrics, it has limited success in passing the DM tests. After conducting a comprehensive assessment of all prediction tasks and models, significant improvements over the LR model can be

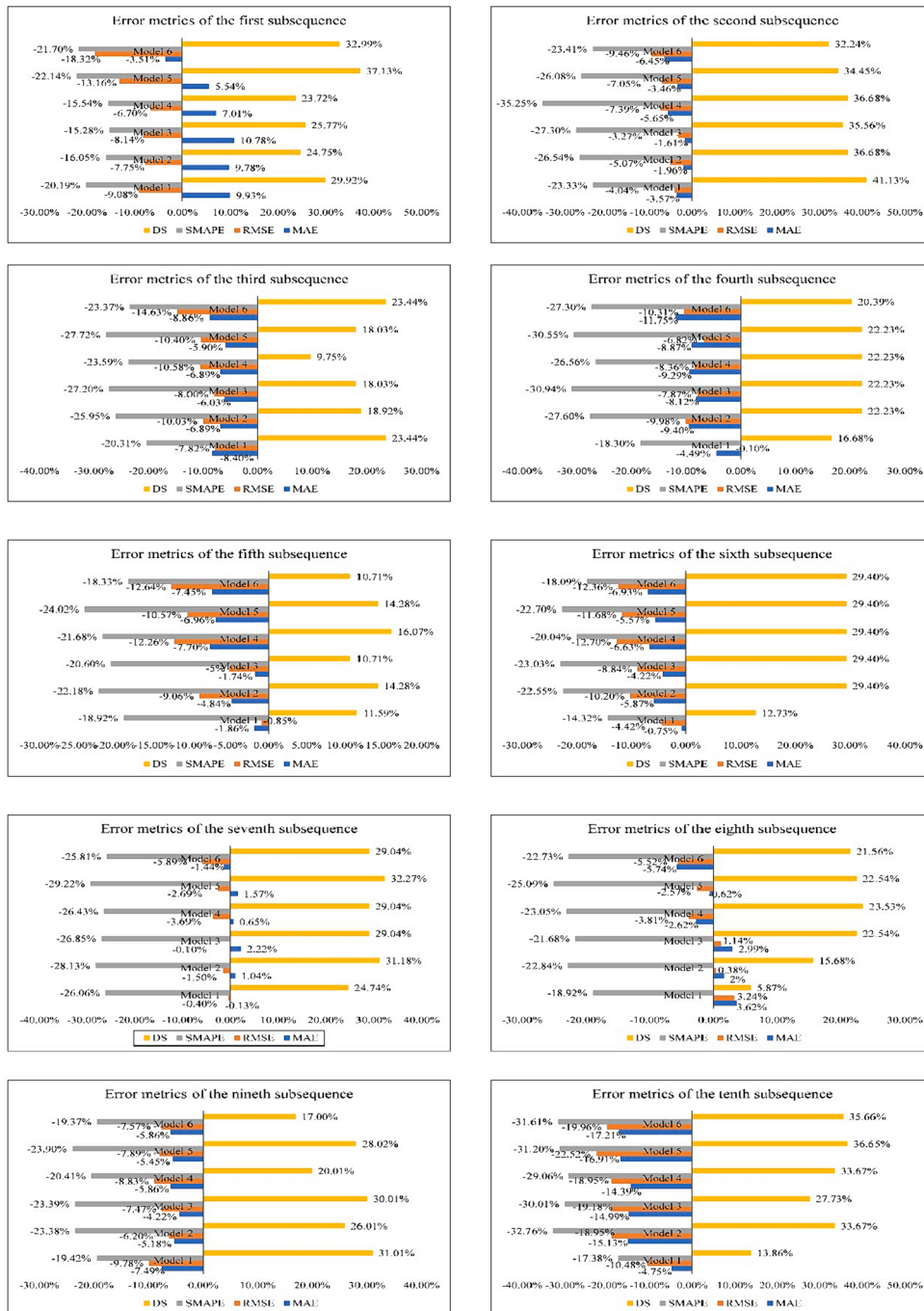


Fig. 11. Error metrics results of ten subsequences upon the “MF-EMD-ML” system **Note:** Model one to Model six respectively represent MF-EMD-SVR, MF-EMD-RF, MF-EMD-XGB, MF-EMD-LSTM, MF-EMD-GRU, and MF-EMD-CNN.

observed only in the following instances:

1. The MF-EMD-SVR and MF-EMD-CNN models in the third subsequence prediction.
2. The MF-EMD-RF, MF-EMD-LSTM, and MF-EMD-CNN models in the fourth prediction task.
3. The MF-EMD-CNN model in the fifth subsequence prediction.
4. The MF-EMD-SVR model in the ninth prediction task.
5. The MF-EMD-RF, MF-EMD-XGB, MF-EMD-LSTM, MF-EMD-GRU, and MF-EMD-CNN models in the tenth prediction task.

In all other instances, the DM test results fails to produce statistically significant outcomes. This implies that the effectiveness of the proposed “Mixed-Frequency-Rolling Decomposition-Prediction” framework requires further validation.

4.2.2. Prediction results of the “MF-EEMD-ML” system

In Section 4.2.1, we utilize the Empirical Mode Decomposition (EMD) algorithm, which enhances predictive performance but has

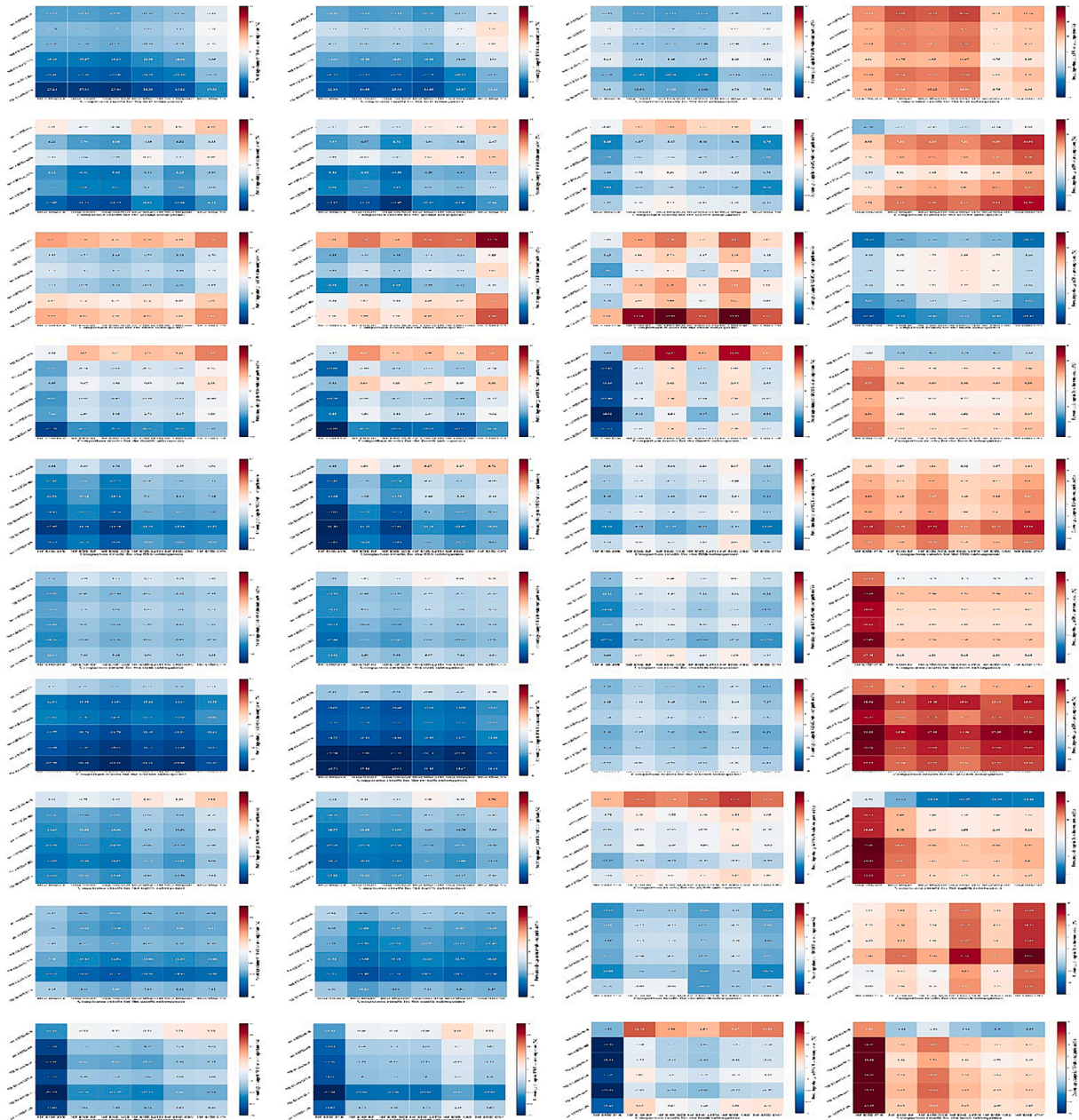


Fig. 12. The extent of prediction enhancement offered by “MF-EEMD-ML” compared with “MF-EMD-ML”. (If Fig. 11 is not clear, readers can download supplementary material for examination.) Note: Each heatmap figure shows the forecasting improvement extent of specific indicator.

limited success in passing the Diebold-Mariano (DM) test. In this section, we conduct further investigations by replacing EMD with the Ensemble Empirical Mode Decomposition (EEMD) algorithm. This is done to validate the effectiveness of the “MF-EEMD-ML” prediction system. [Appendix Table A3](#) shows the horizontal and vertical accuracy metrics for the predictions of six models. Compared to the results from the “MF-EMD-ML” framework in [Table A2](#), most metrics in [Table A3](#) exhibit improvements. Since all models in [Table A2](#) outperform the LR model, we do not analyze their improvements relative to LR. Instead, our focus is on evaluating the enhancement over the “MF-EMD-ML” system presented in [Section 4.2.1](#). The relevant outcomes are depicted in [Fig. 12](#).

In [Fig. 12](#), there are four heatmaps in each row, corresponding to four metrics of improvement for each of the ten subsequence prediction tasks. The vertical axis of each heatmap represents six models within the “MF-EEMD-ML” prediction framework, while the horizontal axis depicts six models under the “MF-EMD-ML” framework. To indicate the extent of model improvement, we calculate the rate of change in error metrics for models on the vertical axis relative to those on the horizontal axis, as shown by the depth of color in the heatmaps. For the MAE, RMSE, and SMAPE metrics, deeper shades of blue signify greater predictive accuracy improvement. In contrast, for the DS metric, deeper shades of red represent a higher enhancement in model precision. It is apparent that, except for the prediction of the third stock return subsequence, the “MF-EEMD-ML” framework significantly enhances performance over the “MF-EMD-ML” framework. For example, in the first prediction task, the MF-EEMD-CNN model achieves a 27.80 % improvement in MAE, a 23.68 % improvement in RMSE, a 14.33 % improvement in SMAPE, and a 13.12 % enhancement in DS compared to the MF-EMD-XGB model.

Furthermore, the diagonal of each heatmap in [Fig. 12](#) shows the improvement extent in the “MF-EEMD-ML” prediction framework, with the machine learning algorithm remaining unchanged. Statistical analysis reveals the most notable improvements as follows: In the prediction of the first stock return subsequence, MF-EEMD-GRU achieves a 19.18 % reduction in the MAE metric, and MF-EEMD-XGB sees an 11.71 % decrease in the SMAPE metric. For the seventh subsequence, MF-EEMD-GRU demonstrates a 19.08 % reduction in the RMSE metric. In the ninth prediction task, MF-EEMD-LSTM shows a 16.66 % increase in the DS metric. [Fig. 13](#) illustrates the predictive trends for the ten stock returns subsequences using the six models following the reverse mixed-frequency treatment. Compared to the results from the non-rolling decomposition approach in [Section 4.1](#), the “MF-EEMD-ML” system demonstrates significant improvements.

This study validates the economic significance of real-time forum sentiment in predicting stock returns and integrates RR-MIDAS with rolling decomposition algorithms. Experimental results demonstrate a significant enhancement in predictive accuracy with the introduction of the decomposition algorithm. In similar studies, [Lv, Shu, Xu, and Wu \(2022\)](#) forecast stock trends in six markets, showing that the CEEMDAN-DAE-RNN model improves by 7.44 %, 6.25 %, and 19.66 % in RMSE, MAE, and NMSE indicators compared to the benchmark model. [Park, Kim, and Kim \(2022\)](#), when predicting returns for S&P500, SSE Composite Index, and KOSPI200, observe that LSTM-Forest with multi-task (LFM) exhibits RMSE values 25.53 %, 22.75 %, and 16.29 % lower than the benchmark RF model. In our research, the “MF-EEMD-ML” system surpasses the “MF-EMD-ML” system, with the maximum improvement reaching 19.08 %, 19.18 %, and 11.71 % in RMSE, MAE, and SMAPE indicators. These improvements are substantial compared to [Lv et al. \(2022\)](#). Given that [Park et al. \(2022\)](#) focus on predicting the direction of stock movements, a task simpler than predicting numerical returns, the higher improvement is expected. Additionally, while many other stocks market prediction studies using decomposition algorithms claim higher predictive accuracy, they often suffer from data leakage issues, making it challenging to assess their practical application.

In this section, we conduct two sets of DM tests. The first set aims to verify whether the “MF-EEMD-ML” prediction system significantly outperforms the LR model, while the second set is intended to ascertain whether the “MF-EEMD-ML” system shows significant superiority over the “MF-EMD-ML” system. The respective results are presented in [Appendix Tables A.4 and A.5](#). An examination of the test outcomes in [Table A4](#) reveals that the MF-EEMD-SVR, MF-EEMD-RF, and MF-EEMD-XGB models do not surpass the LR model in predicting the first subsequence. Similarly, for the second subsequence prediction, the MF-EEMD-SVR and MF-EEMD-XGB models, and for the third subsequence prediction, the MF-EEMD-SVR, MF-EEMD-GRU, and MF-EEMD-CNN models, as well as the MF-EEMD-SVR model in the fourth, seventh, and eighth subsequences, fail to outperform the LR model. However, in all other cases, the DM tests yield statistically significant results. This indicates that the proposed “MF-EEMD-ML” prediction system effectively enhances the accuracy of high-frequency stock returns predictions.

The results presented in [Table A5](#) show that, except for the second and third stock return subsequences, the “MF-EEMD-ML” prediction system successfully passes the DM tests when compared to the “MF-EMD-ML” system. Coupled with the bar charts in [Fig. 11](#), which illustrate the degree of improvement in various indicators, we can infer that for the task of reverse mixed-frequency prediction of 3-minute stock returns, the “MF-EEMD-ML” system demonstrates significant superiority over the “MF-EMD-ML” system.

Regarding the trade-off between model computational efficiency and prediction accuracy, this study tackles the task by dividing it into ten models for estimation after addressing the mixed-frequency data relationship between sentiment and stock returns. The processes of decomposition, hyperparameter selection, fitting, and obtaining one-step-ahead forecasting values for each model take only a few minutes. The simultaneous execution of six machine learning algorithms enables the direct derivation of predictions for 10 consecutive 3-minute stock returns over the next 30 min, achieving the predictive accuracy presented in this study. Unlike composite models that are time-consuming, the machine learning algorithms in this research do not consume much time. We also incorporate early stopping to prevent overfitting of the neural networks. Therefore, computational efficiency is not the primary concern of this study.

The calculation equation is:

$$Indicator(\%) = 100 \times (Indicator_{inrows} - Indicator_{incolumns}) / (Indicator_{inrows})$$

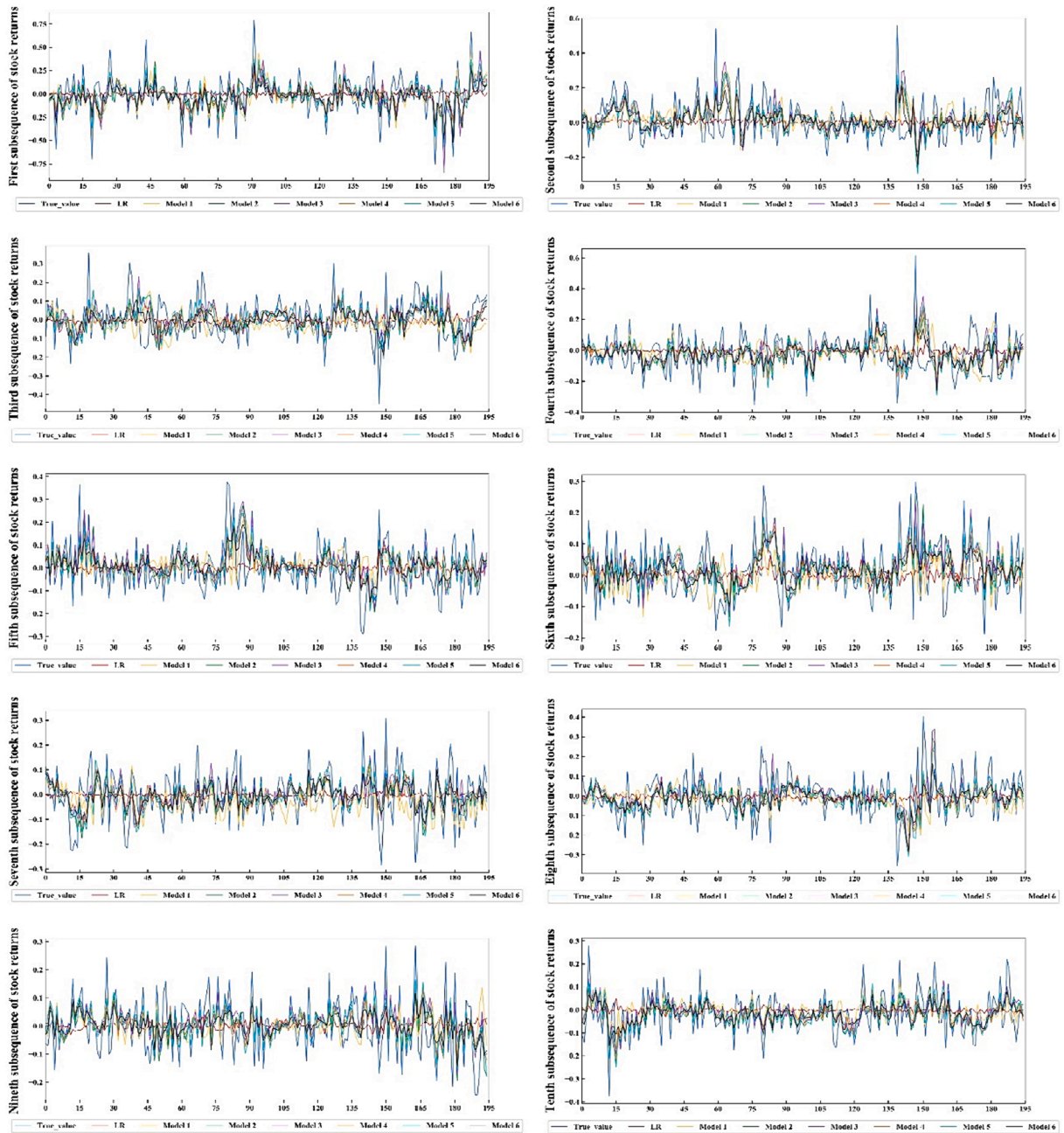


Fig. 13. Predictive trends of the ten stock returns subsequences for the six models **Note:** Model one to Model six respectively represent MF-EEMD-SVR, MF-EEMD-RF, MF-EEMD-XGB, MF-EEMD-LSTM, MF-EEMD-GRU, and MF-EEMD-CNN.

5. Conclusions

This paper explores the significant influence of real-time investor sentiment from stock message boards on high-frequency stock returns. It also introduces a “Mixed-Frequency-Rolling Decomposition-Forecasting” framework to improve the accuracy of high-frequency stock returns predictions. Economic analysis using the RR-MIDAS model reveals that while the lagged period of return itself has limited impact, half-hour frequency sentiment from stock message boards can affect 3-minute stock returns over seven high-frequency periods. Notably, this influence shows a trend of diminishing impact during periods of volatility. It is noteworthy that this influence is volatile and unstable, and it exhibits a gradually diminishing trend.

On the other hand, the outcomes from [Section 4.1](#) indicate that due to the high noise and volatility clustering characteristics of the

stock returns, even with the application of machine learning algorithms coupled with half-hourly sentiment from stock message boards, achieving the desired predictive efficacy remains challenging. To address this, we introduce the “Mixed-Frequency-Rolling Decomposition-Prediction” framework. Inspired by Foroni’s (2018) reverse mixed-frequency data sampling approach, we divide the 3-minute frequency stock returns sequence into ten subsequences, each corresponding to a different time period. Using a fixed window, each subsequence undergoes a “rolling decomposition - one-step-ahead prediction” process. We compute fuzzy entropy values for each component resulting from the decomposition, and then reconstruct these components into high-frequency, mid-frequency, and low-frequency components. Pearson correlation coefficients reveal the lowest autocorrelation in the high-frequency component. Consequently, predictions are made by combining this component with half-hourly sentiment from stock message boards. The results from the high-frequency, mid-frequency, and low-frequency components are then aggregated to obtain the predictive values. To ensure the stability of these outcomes, six classical machine learning algorithms are utilized.

In the initial attempt within the “MF-EMD-ML” framework, the six machine learning algorithms exhibit improvements over the conventional LR model in MAE, RMSE, SMAPE, and DS metrics. However, the Diebold-Mariano (DM) tests conducted on these prediction outcomes do not meet the significance threshold. Subsequently, employing the EEMD decomposition algorithm yield more successful results. Compared to the “MF-EMD-ML” prediction framework, the “MF-EEMD-ML” system achieves a maximum reduction of 19.18 % in MAE, 19.08 % in RMSE, 11.71 % in SMAPE, and a maximum improvement of 16.66 % in DS. Furthermore, DM test results show that the “MF-EEMD-ML” system not only passes the significance test compared to the LR model but also against the “MF-EMD-ML” framework. This underscores the stability and effectiveness of its predictions.

This study contributes to the understanding of the timeliness and complexity of information structures in the stock market for investors and regulatory authorities. Investors should discern the authenticity of internet information, avoiding being easily swayed by extreme sentiments on forums, and enhances rational investment and decision-making capabilities. Regulatory authorities should incorporate various social media platforms, such as Weibo and forums, into the scope of information supervision. They can make full use of the model proposed in this study to monitor forum sentiments in real-time, mitigating the adverse effects of excessive market downturns.

The limitations of our study lie in not considering the economic impact and predictive effects of overnight sentiment on stock returns, a topic discussed by Sun et al. (2016). In theory, posts issued after the market closes at 15:00 until the opening at 9:30 the next day are considered night posts, with a larger time span. These posts are more dispersed and cannot be concentrated, unlike the posts during trading hours examined in this study. Therefore, future research can extend the modeling in this paper to explore the predictive effects of overnight sentiment on stock returns for the next day, as well as the predictive effects of midday sentiment on afternoon stock returns.

CRedit authorship contribution statement

Yi Cai: Data curation, Formal analysis, Methodology, Software, Writing – original draft. **Zhenpeng Tang:** Funding acquisition, Investigation, Supervision, Validation, Visualization. **Ying Chen:** Investigation, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table A1

Horizontal and vertical error metrics for the MF-EMD-ML prediction system.

Target	Evaluation metrics	LR	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+1/m}$	MAE	0.1299	0.1428	0.1426	0.1439	0.1390	0.1371	0.1255
	RMSE	0.2014	0.1831	0.1858	0.1850	0.1879	0.1749	0.1645

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Table A1 (continued)

Target	Evaluation metrics	LR	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+2/m}$	SMAPE	1.5851	1.2650	1.3307	1.3429	1.3387	1.2342	1.2411
	DS	0.4974	0.6462	0.6205	0.6256	0.6154	0.6821	0.6615
	MAE	0.0868	0.0837	0.0851	0.0854	0.0819	0.0838	0.0812
	RMSE	0.1163	0.1116	0.1104	0.1125	0.1077	0.1081	0.1053
$return_{t+3/m}$	SMAPE	1.7507	1.3423	1.2861	1.2727	1.2944	1.2942	1.3408
	DS	0.4615	0.6513	0.6308	0.6256	0.6308	0.6205	0.6103
	MAE	0.0813	0.0750	0.0757	0.0764	0.0757	0.0765	0.0741
	RMSE	0.1087	0.1002	0.0978	0.1000	0.0972	0.0974	0.0928
$return_{t+4/m}$	SMAPE	1.6365	1.3041	1.2119	1.1913	1.2505	1.1829	1.2541
	DS	0.5692	0.7026	0.6769	0.6718	0.6667	0.6718	0.7026
	MAE	0.0936	0.0894	0.0848	0.0860	0.0849	0.0853	0.0826
	RMSE	0.1232	0.1220	0.1109	0.1135	0.1129	0.1148	0.1105
$return_{t+5/m}$	SMAPE	1.6774	1.3704	1.2144	1.1584	1.2319	1.1649	1.2195
	DS	0.5538	0.6462	0.6769	0.6769	0.6769	0.6769	0.6667
	MAE	0.0805	0.0790	0.0766	0.0791	0.0743	0.0749	0.0745
	RMSE	0.1060	0.1051	0.0964	0.1005	0.0930	0.0948	0.0926
$return_{t+6/m}$	SMAPE	1.6256	1.3181	1.2650	1.2907	1.2731	1.2351	1.3276
	DS	0.5744	0.6410	0.6564	0.6359	0.6667	0.6564	0.6359
	MAE	0.0664	0.0659	0.0625	0.0636	0.0620	0.0627	0.0618
	RMSE	0.0882	0.0843	0.0792	0.0804	0.0770	0.0779	0.0773
$return_{t+7/m}$	SMAPE	1.6027	1.3732	1.2413	1.2336	1.2815	1.2389	1.3127
	DS	0.5231	0.5897	0.6769	0.6769	0.6769	0.6769	0.6769
	MAE	0.0766	0.0765	0.0774	0.0783	0.0771	0.0778	0.0755
	RMSE	0.1002	0.0998	0.0987	0.1001	0.0965	0.0975	0.0943
$return_{t+8/m}$	SMAPE	1.7736	1.3114	1.2746	1.2947	1.3048	1.2553	1.3159
	DS	0.4769	0.5949	0.6256	0.6154	0.6154	0.6308	0.6154
	MAE	0.0802	0.0831	0.0815	0.0826	0.0781	0.0797	0.0756
	RMSE	0.1050	0.1084	0.1054	0.1062	0.1010	0.1023	0.0992
$return_{t+9/m}$	SMAPE	1.6756	1.3586	1.2899	1.3124	1.2893	1.2552	1.2947
	DS	0.5231	0.5538	0.6051	0.6410	0.6462	0.6410	0.6359
	MAE	0.0734	0.0679	0.0696	0.0703	0.0691	0.0694	0.0691
	RMSE	0.0951	0.0858	0.0892	0.0880	0.0867	0.0876	0.0879
$return_{t+10/m}$	SMAPE	1.6744	1.3492	1.2829	1.2828	1.3326	1.2742	1.3500
	DS	0.5128	0.6718	0.6462	0.6667	0.6154	0.6564	0.6000
	MAE	0.0674	0.0642	0.0572	0.0573	0.0577	0.0560	0.0558
	RMSE	0.0897	0.0803	0.0727	0.0725	0.0727	0.0695	0.0718
	SMAPE	1.7415	1.4388	1.1709	1.2188	1.2355	1.1981	1.1910
	DS	0.5179	0.5897	0.6923	0.6615	0.6923	0.7077	0.7026

Note: Model 1 to Model 6 represent MF-EMD-SVR, MF-EMD-RF, MF-EMD-XGB, MF-EMD-LSTM, MF-EMD-GRU, and MF-EMD-CNN, respectively. These definitions apply to Table B as well. Bold entries in the table indicate models with the best performance for individual metrics.

Table A2

DM test for the MF-EMD-ML prediction system.

Target	Target model	Benchmark model					
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+1/m}$	LR	−1.39 (0.16)	−1.48 (0.14)	−1.59 (0.11)	−1.18 (0.24)	−0.76 (0.45)	0.48 (0.63)
	Model 1		−0.08 (0.94)	−0.37 (0.71)	0.53 (0.59)	1.47 (0.14)	4.08*** (4.46e-05)
	Model 2			−0.51 (0.61)	0.61 (0.54)	1.66* (0.10)	5.04*** (4.59e-07)
	Model 3				0.79 (0.43)	1.84* (0.07)	4.91*** (9.14e-07)
	Model 4					0.33 (0.74)	2.35** (0.02)
	Model 5						3.09*** (0.00)
	Model 6						
$return_{t+2/m}$	LR	0.82 (0.41)	0.33 (0.74)	0.26 (0.80)	1.13 (0.26)	0.62 (0.53)	1.43 (0.15)
	Model 1		−0.45 (0.65)	−0.46 (0.65)	0.62 (0.53)	−0.05 (0.96)	0.85 (0.39)
	Model 2			−0.15 (0.88)	1.92* (0.05)	0.72 (0.47)	1.83* (0.07)
	Model 3				1.35 (0.18)	0.61 (0.54)	1.38 (0.17)
	Model 4					−1.45 (0.15)	0.40 (0.69)
	Model 5						1.28 (0.20)
	Model 6						
$return_{t+3/m}$	LR	2.22** (0.03)	1.18 (0.24)	0.98 (0.33)	1.36 (0.17)	0.99 (0.32)	1.87* (0.06)
	Model 1		−0.23 (0.82)	−0.36 (0.72)	−0.22 (0.83)	−0.44 (0.66)	0.33 (0.74)
	Model 2			−0.43 (0.67)	0.10 (0.92)	−0.56 (0.57)	0.92 (0.36)
	Model 3				0.38 (0.71)	−0.08 (0.94)	0.99 (0.32)
	Model 4					−0.67 (0.50)	1.13 (0.26)
							1.34 (0.18)

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Table A2 (continued)

Target	Target model	Benchmark model					
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+4/m}$	Model 5						
	Model 6						
	LR	0.95 (0.34)	1.81* (0.07)	1.48 (0.14)	1.85* (0.06)	1.47 (0.14)	2.45** (0.01)
	Model 1		1.20 (0.23)	0.91 (0.36)	1.42 (0.16)	1.14 (0.26)	2.04** (0.04)
	Model 2			−0.80 (0.42)	−0.04 (0.97)	−0.24 (0.81)	1.06 (0.29)
	Model 3				0.58 (0.56)	0.35 (0.73)	1.47 (0.14)
	Model 4					−0.28 (0.78)	1.74* (0.08)
$return_{t+5/m}$	Model 5						1.25 (0.21)
	Model 6						
	LR	0.47 (0.64)	0.90 (0.37)	0.32 (0.75)	1.54 (0.12)	1.28 (0.20)	1.68* (0.09)
	Model 1		0.74 (0.46)	−0.01 (0.99)	1.57 (0.12)	1.30 (0.19)	1.46 (0.15)
	Model 2			−1.76* (0.08)	1.52 (0.13)	1.11 (0.27)	1.16 (0.25)
	Model 3				2.24** (0.03)	2.14** (0.03)	1.84* (0.07)
	Model 4					−0.46 (0.64)	−0.03 (0.97)
$return_{t+6/m}$	Model 5						0.26 (0.79)
	Model 6						
	LR	0.15 (0.88)	0.91 (0.36)	0.60 (0.55)	1.25 (0.21)	0.85 (0.40)	1.43 (0.15)
	Model 1		1.13 (0.26)	0.66 (0.51)	1.50 (0.13)	1.08 (0.28)	1.52 (0.13)
	Model 2			−1.10 (0.27)	0.30 (0.77)	−0.27 (0.79)	0.34 (0.73)
	Model 3				0.89 (0.37)	0.48 (0.63)	0.84 (0.40)
	Model 4					−0.74 (0.46)	0.17 (0.87)
$return_{t+7/m}$	Model 5						0.62 (0.54)
	Model 6						
	LR	0.01 (0.99)	−0.28 (0.78)	−0.49 (0.62)	−0.24 (0.81)	−0.37 (0.71)	0.21 (0.83)
	Model 1		−0.40 (0.69)	−0.64 (0.52)	−0.34 (0.74)	−0.53 (0.60)	0.25 (0.81)
	Model 2			−0.73 (0.47)	0.18 (0.86)	−0.26 (0.79)	1.12 (0.26)
	Model 3				0.61 (0.54)	0.26 (0.80)	1.38 (0.17)
	Model 4					−0.59 (0.55)	1.24 (0.21)
$return_{t+8/m}$	Model 5						1.35 (0.18)
	Model 6						
	LR	−0.71 (0.48)	−0.29 (0.77)	−0.51 (0.61)	0.61 (0.54)	0.16 (0.87)	1.42 (0.15)
	Model 1		−0.46 (0.65)	0.09 (0.93)	1.73* (0.08)	1.08 (0.28)	2.55** (0.01)
	Model 2			−0.78 (0.44)	2.57** (0.01)	1.45 (0.15)	3.19*** (0.00)
	Model 3				2.23** (0.03)	1.53 (0.13)	2.82*** (0.00)
	Model 4					−1.62 (0.10)	1.88* (0.06)
$return_{t+9/m}$	Model 5						2.36** (0.02)
	Model 6						
	LR	2.02** (0.04)	0.88 (0.38)	0.75 (0.45)	1.22 (0.22)	0.98 (0.33)	1.31 (0.19)
	Model 1		−0.67 (0.50)	−0.91 (0.36)	−0.63 (0.53)	−0.61 (0.54)	−0.56 (0.57)
	Model 2			−0.49 (0.63)	0.31 (0.76)	0.16 (0.87)	0.27 (0.79)0.54
	Model 3				0.65 (0.52)	0.53 (0.60)	(0.59)
	Model 4					−0.23 (0.82)	0.04 (0.97)
$return_{t+10/m}$	Model 5						0.17 (0.87)
	Model 6						
	LR	1.23 (0.22)	2.87*** (0.00)	2.79*** (0.01)	3.10*** (0.00)	3.27*** (0.00)	3.77*** (0.00)
	Model 1		2.47** (0.01)	2.66*** (0.01)	2.70*** (0.01)	3.17*** (0.00)	3.36*** (0.00)
	Model 2			−0.16 (0.88)	−0.36 (0.72)	1.08 (0.28)	1.00 (0.32)
	Model 3				−0.21 (0.83)	1.06 (0.29)	1.00 (0.32)
	Model 4					1.56 (0.12)	1.84* (0.07)
$return_{t+11/m}$	Model 5						0.13 (0.90)
	Model 6						
	LR						
	Model 1						
	Model 2						
	Model 3						
	Model 4						

Note: When the value of the t-statistic is positive and significant, it can be concluded that the baseline model outperforms the target model. The bolded part indicates that the t-statistic is significant.

Table A3

Horizontal and vertical error metrics for the MF-EEMD-ML prediction system.

Target	Evaluation metrics	LR	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+1/m}$	MAE	0.1299	0.1212	0.1253	0.1238	0.1154	0.1108	0.1039
	RMSE	0.2014	0.1568	0.1717	0.1675	0.1563	0.1432	0.1412
	SMAPE	1.5851	1.1290	1.2388	1.1856	1.2240	1.0980	1.1505
	DS	0.4974	0.7385	0.7128	0.7179	0.6872	0.7231	0.7077
$return_{t+2/m}$	MAE	0.0868	0.0846	0.0785	0.0820	0.0769	0.0774	0.0746
	RMSE	0.1163	0.1092	0.1027	0.1095	0.1006	0.1016	0.0969

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Table A3 (continued)

Target	Evaluation metrics	LR	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+3/m}$	SMAPE	1.7507	1.3367	1.2235	1.2389	1.2767	1.2236	1.2810
	DS	0.4615	0.6103	0.6769	0.6667	0.6359	0.6615	0.6821
	MAE	0.0813	0.0797	0.0721	0.0731	0.0719	0.0770	0.0793
	RMSE	0.1087	0.1056	0.0933	0.0953	0.0914	0.0994	0.1015
$return_{t+4/m}$	SMAPE	1.6365	1.2899	1.2359	1.2023	1.2683	1.2459	1.3711
	DS	0.5692	0.6308	0.6769	0.6769	0.6769	0.6462	0.6154
	MAE	0.0936	0.0889	0.0829	0.0844	0.0822	0.0826	0.0778
	RMSE	0.1232	0.1185	0.1088	0.1149	0.1086	0.1103	0.1014
$return_{t+5/m}$	SMAPE	1.6774	1.3035	1.1861	1.1887	1.1994	1.1522	1.1973
	DS	0.5538	0.6410	0.6923	0.7026	0.6821	0.6872	0.6974
	MAE	0.0805	0.0738	0.0692	0.0688	0.0676	0.0648	0.0668
	RMSE	0.1060	0.0979	0.0889	0.0897	0.0851	0.0825	0.0846
$return_{t+6/m}$	SMAPE	1.6256	1.2471	1.2297	1.2062	1.2399	1.1543	1.2599
	DS	0.5744	0.6667	0.6872	0.6974	0.6923	0.7333	0.7026
	MAE	0.0664	0.0597	0.0570	0.0579	0.0571	0.0565	0.0577
	RMSE	0.0882	0.0779	0.0722	0.0724	0.0710	0.0696	0.0724
$return_{t+7/m}$	SMAPE	1.6027	1.2395	1.1904	1.1694	1.2111	1.1328	1.2558
	DS	0.5231	0.6718	0.7282	0.7077	0.7026	0.7282	0.6923
	MAE	0.0766	0.0727	0.0666	0.0673	0.0652	0.0642	0.0631
	RMSE	0.1002	0.0928	0.0838	0.0844	0.0831	0.0789	0.0791
$return_{t+8/m}$	SMAPE	1.7736	1.2026	1.2216	1.2374	1.1934	1.2023	1.2364
	DS	0.4769	0.6615	0.7077	0.6923	0.7231	0.7077	0.6974
	MAE	0.0802	0.0801	0.0720	0.0713	0.0704	0.0689	0.0711
	RMSE	0.1050	0.1017	0.0928	0.0913	0.0893	0.0881	0.0919
$return_{t+9/m}$	SMAPE	1.6756	1.4731	1.3081	1.2776	1.2828	1.2055	1.3186
	DS	0.5231	0.5385	0.6564	0.6564	0.6923	0.6872	0.6769
	MAE	0.0734	0.0632	0.0626	0.0635	0.0615	0.0594	0.0637
	RMSE	0.0951	0.0813	0.0788	0.0778	0.0764	0.0745	0.0801
$return_{t+10/m}$	SMAPE	1.6744	1.1982	1.2219	1.2298	1.2208	1.1916	1.2715
	DS	0.5128	0.6821	0.6872	0.6872	0.7179	0.6667	0.6718
	MAE	0.0674	0.0570	0.0530	0.0518	0.0532	0.0507	0.0529
	RMSE	0.0897	0.0720	0.0683	0.0674	0.0672	0.0626	0.0678
	SMAPE	1.7415	1.3163	1.1512	1.1504	1.1892	1.1366	1.2197
	DS	0.5179	0.6462	0.7333	0.7128	0.7333	0.7385	0.7282

Note: Model 1 to Model 6 represent MF-EEMD-SVR, MF-EEMD-RF, MF-EEMD-XGB, MF-EEMD-LSTM, MF-EEMD-GRU, and MF-EEMD-CNN, respectively. These definitions apply to Table D and Table E as well. Bold entries in the table indicate models with the best performance for individual metrics.

Table A4

DM test for the MF-EEMD-ML prediction system (regarding the LR model).

Target	Target model	Benchmark model					
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+1/m}$	LR	0.76 (0.45)	0.45 (0.65)	0.61 (0.54)	1.84* (0.07)	2.01** (0.04)	3.38*** (0.00)
	Model 1		−0.82 (0.41)	−0.57 (0.57)	1.13 (0.26)	2.60*** (0.01)	3.28*** (0.00)
	Model 2			0.49 (0.63)	2.73*** (0.01)	3.21*** (0.00)	4.72*** (2.38e-06)
	Model 3				2.25** (0.02)	3.12*** (0.00)	4.16*** (3.12e-05)
	Model 4					1.25 (0.21)	2.95*** (0.00)
	Model 5						1.63 (0.10)
$return_{t+2/m}$	LR	0.51 (0.61)	1.75* (0.08)	0.94 (0.35)	2.36** (0.02)	1.90* (0.06)	3.22*** (0.00)
	Model 1		1.66* (0.10)	0.73 (0.47)	2.50** (0.01)	2.30** (0.02)	3.04*** (0.00)
	Model 2			−2.04** (0.04)	0.96 (0.34)	0.61 (0.54)	1.62 (0.11)
	Model 3				2.09** (0.04)	2.00** (0.04)	2.49** (0.01)
	Model 4					−0.33 (0.74)	1.14 (0.25)
	Model 5						1.11 (0.27)
$return_{t+3/m}$	LR	0.56 (0.58)	2.35** (0.02)	2.02** (0.04)	2.64*** (0.01)	0.99 (0.32)	0.57 (0.57)
	Model 1		2.01** (0.04)	1.63 (0.10)	2.30** (0.02)	0.63 (0.53)	0.06 (0.95)
	Model 2			−0.63 (0.53)	0.14 (0.89)	−2.41** (0.02)	−2.66*** (0.01)
	Model 3				0.69 (0.49)	−1.73* (0.08)	−2.05** (0.04)
	Model 4					−2.81*** (0.00)	−2.90*** (0.00)
	Model 5						−0.70 (0.48)

(continued on next page)

Table A4 (continued)

Target	Target model	Benchmark model					
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+4/m}$	LR	0.95 (0.34)	2.39** (0.02)	1.79* (0.07)	2.66*** (0.01)	2.15** (0.03)	4.06*** (4.95e-05)
	Model 1		1.57 (0.12)	1.17 (0.24)	1.79* (0.07)	1.78* (0.04)	3.05*** (0.00)
	Model 2			−0.74 (0.46)	0.32 (0.75)	0.08 (0.94)	2.52** (0.01)
	Model 3				0.88 (0.38)	0.74 (0.46)	2.39** (0.02)
	Model 4					−0.22 (0.82)	2.31** (0.02)
	Model 5						1.97** (0.05)
$return_{t+5/m}$	LR	2.02** (0.04)	2.68*** (0.01)	2.65*** (0.01)	3.53*** (0.00)	3.81*** (0.00)	4.22*** (2.42e-05)
	Model 1		1.47 (0.14)	1.72* (0.09)	2.33** (0.02)	3.34*** (0.00)	2.46** (0.01)
	Model 2			0.33 (0.74)	0.94 (0.35)	2.77*** (0.01)	1.08 (0.28)
	Model 3				0.62 (0.54)	2.38** (0.02)	0.78 (0.43)
	Model 4					2.16** (0.03)	0.52 (0.60)
	Model 5						−0.99 (0.32)
$return_{t+6/m}$	LR	2.20** (0.03)	2.52** (0.01)	2.20** (0.03)	2.79*** (0.01)	2.62*** (0.01)	3.05*** (0.00)
	Model 1		0.84 (0.40)	0.48 (0.63)	0.86 (0.39)	1.07 (0.28)	0.66 (0.51)
	Model 2			−0.80 (0.43)	−0.10 (0.92)	0.27 (0.79)	−0.35 (0.73)
	Model 3				0.52 (0.60)	0.90 (0.37)	0.18 (0.86)
	Model 4					0.52 (0.60)	−0.34 (0.74)
	Model 5						−0.58 (0.56)
$return_{t+7/m}$	LR	1.02 (0.31)	2.97*** (0.00)	2.80*** (0.01)	3.64*** (0.00)	3.48*** (0.00)	4.72*** (2.41e-06)
	Model 1		1.77* (0.08)	1.55 (0.12)	2.38** (0.02)	2.62*** (0.01)	3.07*** (0.00)
	Model 2			−0.48 (0.63)	0.84 (0.40)	1.40 (0.16)	1.95* (0.05)
	Model 3				0.16 (0.25)	1.67* (0.10)	2.32** (0.02)
	Model 4					0.76 (0.45)	1.32 (0.19)
	Model 5						0.65 (0.51)
$return_{t+8/m}$	LR	0.09 (0.93)	2.37** (0.02)	2.56** (0.01)	3.06*** (0.00)	3.15*** (0.00)	3.12*** (0.00)
	Model 1		2.52** (0.01)	2.89*** (0.00)	3.48*** (0.00)	3.85*** (0.00)	2.82*** (0.00)
	Model 2			0.42 (0.68)	1.01 (0.31)	1.88* (0.06)	0.46 (0.65)
	Model 3				0.58 (0.56)	1.37 (0.17)	0.14 (0.89)
	Model 4					1.12 (0.26)	−0.47 (0.64)
	Model 5						−1.10 (0.27)
$return_{t+9/m}$	LR	3.52*** (0.00)	3.16*** (0.00)	2.91*** (0.00)	4.05*** (5.17e-05)	3.96*** (7.36e-05)	3.63*** (0.00)
	Model 1		0.24 (0.81)	−0.04 (0.97)	0.84 (0.40)	1.60 (0.11)	−0.23 (0.82)
	Model 2			−0.48 (0.63)	0.82 (0.41)	2.01** (0.04)	−0.57 (0.57)
	Model 3				1.17 (0.24)	2.26** (0.02)	−0.20 (0.84)
	Model 4					1.62 (0.11)	−1.52 (0.13)
	Model 5						−2.22** (0.03)
$return_{t+10/m}$	LR	3.78*** (0.00)	4.20*** (2.63e-05)	4.55*** (5.28e-06)	4.45*** (8.67e-06)	4.74*** (2.13e-06)	5.18*** (2.16e-07)
	Model 1		1.39 (0.16)	2.02** (0.04)	1.51 (0.13)	2.47** (0.01)	1.72* (0.09)
	Model 2			1.03 (0.30)	−0.13 (0.89)	1.46 (0.14)	0.11 (0.91)
	Model 3				−0.98 (0.33)	0.63 (0.53)	−0.66 (0.51)
	Model 4					2.00** (0.05)	0.26 (0.79)
	Model 5						−1.35 (0.18)

Table A5

DM test for the MF-EEMD-ML prediction system (regarding the MF-EMD-ML framework).

Target	Benchmark model					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
$return_{t+1/m}$	2.91*** (0.00)	2.42** (0.02)	2.54** (0.01)	3.25*** (0.00)	3.62*** (0.00)	3.33*** (0.00)
$return_{t+2/m}$	−0.36 (0.72)	2.23** (0.03)	1.04 (0.30)	2.02** (0.04)	2.27** (0.02)	2.71*** (0.01)
$return_{t+3/m}$	−1.96** (0.05)	1.64 (0.10)	1.32 (0.19)	1.86* (0.06)	−0.19 (0.85)	−1.75* (0.08)
$return_{t+4/m}$	0.14 (0.89)	0.69 (0.49)	0.49 (0.62)	0.93 (0.35)	0.81 (0.42)	1.74 (0.08)
$return_{t+5/m}$	2.27** (0.02)	3.01*** (0.00)	3.67*** (0.00)	3.11*** (0.00)	4.06*** (4.84e-05)	3.68*** (0.00)
$return_{t+6/m}$	3.34*** (0.00)	2.62*** (0.01)	2.52** (0.01)	2.60*** (0.01)	2.85*** (0.00)	2.26** (0.02)
$return_{t+7/m}$	1.46 (0.14)	4.49*** (7.27e-06)	4.12*** (3.83e-05)	5.43*** (5.57e-08)	5.50*** (3.77e-08)	5.31*** (1.09e-07)
$return_{t+8/m}$	1.01 (0.31)	3.49*** (0.00)	3.67*** (0.00)	3.08*** (0.00)	4.17*** (3.04e-05)	2.09** (0.04)
$return_{t+9/m}$	2.32** (0.02)	2.79*** (0.01)	2.46** (0.01)	3.51*** (0.00)	4.22*** (2.47e-05)	2.37** (0.02)
$return_{t+10/m}$	4.29*** (1.82e-05)	2.33** (0.02)	2.79*** (0.01)	2.54** (0.01)	2.81*** (0.00)	1.90* (0.06)

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.najef.2024.102147>.

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